Bird_Call_Report_Final

May 23, 2025

1 DASC 400 Bird Call Project

1.0.1 Breanna Ranglall & Ibuki Washburn

[1]:
```
# run these if don't have them installed
#!pip install optuna
#!pip install tensorflow
```

[7]:
```
# standard libraries
import numpy as np
import pandas as pd
import warnings
warnings.simplefilter('ignore')
import gc

# Visualize training history
import matplotlib.pyplot as plt
%matplotlib inline

# tensorflow
import tensorflow as tf
from tensorflow import keras
from tqdm.keras import TqdmCallback

# optuna
import optuna

# sklearn
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
```

1.1 Introduction & Motivation

Automated recognition of bird species from their vocalisations has become an attractive research problem because it blends ecological monitoring with modern machine-learning techniques. Inspired by Cornell Lab's highly regarded Birdsong Recognition competition, we set out to build a classifier that can identify species using only short audio clips. Besides heeding Dr Werness's suggestion that this would make an engaging final project, we were eager to gain hands-on experience with audio

feature engineering—a skill that transfers to many domains, from speech analytics to bioacoustics.

1.2 Exploratory Data Analysis & Project Goals

Preliminary EDA confirmed that class imbalance is substantial, several rare species appear fewer than ten times, making naïve accuracy misleading. Accordingly, our four goals were:

1. Understand the latent structure of the feature space (clusters or outliers).
2. Prototype at least one traditional and two neural models.
3. Tune hyperparameters systematically to squeeze out extra accuracy.
4. Evaluate fairly on a held-out test set and document shortcomings for future work.

[8]:
```
df = pd.read_csv('dataset.csv')

# Drop filename column
df = df.drop(columns=['filename'])

# Convert columns with bird names to numerical code
le = LabelEncoder()
df['label'] = le.fit_transform(df['label'])

df.groupby('label').filter(lambda x : len(x)>10)
sub_df = df[df.groupby('label').label.transform('count')>10].copy()

# view
sub_df
```

[8]:
```
        chroma_stft       rms  spectral_centroid  spectral_bandwidth  \
0          0.592959  0.008214        2756.314965         2671.700479
1          0.445633  0.011182        4849.245849         1960.078954
2          0.491342  0.007226        4048.471445         2235.524317
3          0.555491  0.016315        4327.569353         2846.678935
4          0.687489  0.071456        3634.945700         2473.898862
…             …       …                 …                  …
12360      0.422929  0.032790        2830.771223         2640.725111
12361      0.456022  0.038203        1454.968344         1755.409715
12362      0.541382  0.059542        2967.904198         1894.794915
12363      0.640628  0.023386        1156.133090         2069.722985
12364      0.401658  0.023715        2218.130464         1975.610323

            rolloff  zero_crossing_rate       mfcc1       mfcc2       mfcc3  \
0      5803.582764            0.150738 -329.593048   77.796181  -11.007781
1      6962.341233            0.408716 -506.991882  -94.189995  -66.601006
2      6557.677013            0.302023 -477.984528 -121.129715 -145.713058
3      7981.648952            0.347925 -315.947571   17.248848  -24.313393
4      6342.327588            0.284439  -72.725365   33.178391  -35.684204
…             …                 …          …          …          …
```

```
12360  5853.490556            0.141361 -254.950470   66.365486  -20.901850
12361  2565.192060            0.065938 -262.680817  129.147415  -41.884510
12362  5064.919373            0.205536 -159.406158    3.195026 -129.160080
12363  1551.713326            0.038406 -312.113617  158.666107   42.628853
12364  3853.435032            0.157070 -319.204803  107.682808  -31.604698

           mfcc4  …     mfcc12     mfcc13     mfcc14     mfcc15     mfcc16  \
0      15.441154  …   4.675884  -4.270268   6.639040  -2.032491   3.748571
1      21.418806  …   1.026675  -3.219575   5.016063  -2.692930   1.245947
2     -26.852354  …   0.306044  -4.073335   1.470622  -2.557777   2.980418
3       8.431483  …  -0.374923  -6.571611   2.610246  -5.446579  -3.616137
4      21.969164  …   5.084502   1.275231   4.249923  -4.074911   2.886948
…          …  …        …         …         …         …         …
12360  -6.480114  … -12.614041 -19.589306 -11.288402 -13.606933  -7.365254
12361   2.371792  …   9.449895   6.690379  13.339852   9.990253  14.080956
12362 -20.913094  …   5.464991   2.559317  11.329490   4.002252   6.594406
12363  -6.432523  …   5.457233  -2.644947   1.938845  -1.334749   2.063163
12364   9.592651  …   4.357146  -3.510907   1.607527   1.274850   2.399644

          mfcc17     mfcc18    mfcc19    mfcc20  label
0      -1.192944   3.819380 -1.592449  4.146574      0
1      -2.166367   2.367208 -3.187211  3.042730      0
2       1.284922   5.729064 -2.983708  1.832972      0
3      -4.213018  -1.368571 -5.277973  0.475844      0
4      -6.306517   4.148955 -5.137774  5.145658      0
…          …        …       …        …    …
12360 -12.567055  -6.225912 -9.425448 -1.914419    147
12361   5.101741  11.511769  3.362309  7.456527    147
12362   0.244300   3.474930 -2.202893  5.819273    147
12363  -4.875724   1.170730 -3.790732  1.553186    147
12364  -6.672645  -3.110505 -2.316219  1.500105    147

[12365 rows x 27 columns]
```

1.3 Bird Brain Background and Dataset

Rather than processing the original gigabytes of MP3 files, we leveraged the birdcalldataset Kaggle subset, which already converts each recording into a single row of tabular features. This document summarizes background research on the variables in this dataset:
https://www.kaggle.com/datasets/ladanova/birdcalldataset/data
(based on the competition dataset: https://www.kaggle.com/c/birdsong-recognition/data)

1.3.1 Variables

- `chroma_stft` (Chroma Short-Time Fourier Transform) – spectral
- `rms` (Root Mean Square Energy) – temporal

- `spectral_centroid` – spectral
- `spectral_bandwidth` – spectral
- `rolloff` – spectral
- `zero_crossing_rate` – temporal
- `mfcc1` through `mfcc20` (Mel-Frequency Cepstral Coefficients) – unique

---

1.3.2 Chroma_STFT (Chroma Short-Time Fourier Transform)

Source: - https://www.sciencedirect.com/topics/engineering/continuous-time-fourier-transform - https://www.audiolabs-erlangen.de/content/05_fau/professor/00_mueller/02_teaching/2019w_mpa/2019_Muel AudioFeatures_6.pdf

This is a 12-dimensional representation showing the energy across 12 pitch classes (C, C#, ..., B), **independent of octave**.

Mathematical Steps:

1. Create a time window (possibly 1 ms).
2. Apply a smoothing function.
3. Apply the **Discrete-Time Fourier Transform (DTFT)** or **Fast Fourier Transform (FFT)** to get a frequency spectrum.
4. Use chroma binning on a log scale to reduce to 12 pitch class bins (e.g., A , C).
5. Normalize the resulting vector.

→ If we check our dataset and see just 1 number, they probably summarized the chroma vector using: - Mean value, - Index of dominant pitch class, - Or total chroma energy (if unnormalized).

1.3.3 RMS (Root Mean Square Energy)

Source: https://unison.audio/what-is-rms-in-audio/

- Measures the **average power** of the audio signal over time.

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} x[n]^2}$$

- (x[n]) is a digital sample (also written as (x_n)).
- So: (x[n]^2 = x_n^2)
- In formulas:
  If (x[n]) is a signal in discrete time, then RMS is based on (x[n]^2).

1.3.4 Spectral Centroid

Source: https://www.mathworks.com/help/audio/ug/spectral-descriptors.html

- Represents the **center of mass** of the spectrum.
- Related to the **brightness** of sound: higher frequencies → higher centroid.
- In equations:
- Centroid = $\frac{\sum_k f[k] \cdot S[k]}{\sum_k S[k]}$

  where (f) is frequency, (S) is magnitude. Averaged across the frequency bins.

1.3.5 Spectral Bandwidth (Spectral Spread)

Source: https://www.mathworks.com/help/audio/ug/spectral-descriptors.html

- Measures how spread-out (or the variance of) the frequencies are around the **spectral centroid**.
- Equation resembles the **standard deviation** formula of the spectrum around the centroid.

$\mu_2 = \frac{\sum_{k=b_1}^{b_2} (f_k - \mu_1)^2 \cdot s_k}{\sum_{k=b_1}^{b_2} s_k}$

Where:

- (f_k): frequency in Hz for bin (k)
- (s_k): spectral value at bin (k)
- (b_1, b_2): start and end bin indices
- (μ_1): spectral centroid

1.3.6 Spectral Rolloff

Source: https://www.mathworks.com/help/audio/ug/spectral-descriptors.html

- Indicates **what frequency** contains a given percentage (k) of the total spectral energy.
- Used to estimate **bandwidth concentration** toward lower or higher frequencies.

Rolloff Point = i such that $\sum k = 1\hat{}\{i\}|s_k| = k \sum\{k = 1\}\hat{}\{b_2\}s_k$

Where:

- (s_k): spectral value at bin (k)
- (b_1), (b_2): the lower and upper frequency bins considered
- (k): the chosen energy threshold (typically 85% or 95%)
- (i): the rolloff bin index (converted to Hz later)

Interpretation: This is the frequency below which a certain percentage (k) of the signal energy is concentrated — i.e., how much energy is packed into the lower frequencies.

1.3.7 Zero Crossing Rate

Source: https://medium.com/data-science/decoding-the-symphony-of-sound-audio-signal-processing-for-musical-engineering-c66f09a4d0f5

- Number of times the audio waveform **crosses the zero axis** (time domain).
- For frame (k), ZCR is defined mathematically as the count of sign changes.

ZCR$k =$ $\{i = kK\}$^$\{(k+1)K - 1\}\frac{1}{2}$ |sgn(x_i) − sgn(x_$\{i+1\}$) |

With the sign function:

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Where:

- (K): frame size
- (x_i): audio signal sample at index (i)
-

1.4 The ZCR measures how often the signal crosses zero in a frame.

1.4.1 MFCCs (Mel-Frequency Cepstral Coefficients)

Source: https://www.geeksforgeeks.org/mel-frequency-cepstral-coefficients-mfcc-for-speech-recognition/

Includes: `mfcc1` through `mfcc20`.

- "The Mel-scale is specifically designed to mimic the way humans perceive sound, particularly how we discern differences in pitch. Human hearing is more sensitive to changes in lower frequencies than to equivalent changes in higher frequencies."
- "The Mel-scale addresses this by applying a linear frequency spacing below 1000 Hz and a logarithmic spacing above 1000 Hz. This scaling allows for a more perceptually relevant representation of audio signals, aligning the scale with the non-linear human auditory system:"

How to Compute MFCC? Finally, by taking the first few coefficients from the DCT output, we obtain the MFCCs, which represent a compact and informative description of the speech signal in each frame.
To calculate MFCCs, we follow these steps:

1. **Pre-emphasize the signal**: Amplify higher frequencies to balance the spectrum.

2. **Framing**: Break the signal into small, overlapping frames.

3. **Windowing**: To soften the edges of each frame, apply a Hamming window.

4. **FFT**: Convert each frame from the time domain to the frequency domain.

5. **Mel-filterbank**: Apply overlapping triangular filters spaced according to the Mel-scale.

6. **Logarithm**: To replicate how a human ear reacts to sound strength, take the logarithm of the filterbank outputs.

7. **DCT**: Apply the Discrete Cosine Transform (DCT) to the log Mel-spectrum to obtain the **Mel-frequency Cepstral Coefficients**.

1.4.2 Calculating MFCCs from Speech Signal in Python

In this example, we'll go over how to use Python to calculate the MFCCs from a speech signal. Common libraries like `librosa` for audio processing and `numpy`, `scipy`, and `matplotlib` will be used. Lastly, we'll utilize `ipywidgets` to build a basic GUI that will allow users to test the model in real time.

> **Pipeline**:
> *Original Signal → Pre-emphasis → Framing → Windowing → FFT → Mel-filterbank → Logarithm → DCT → MFCCs*

1.4.3 Additional Sources and Notes

- https://ravinkumar.com/GenAiGuidebook/audio/audio_feature_extraction.html

- https://citeseerx.ist.psu.edu/document?doi=30a90af7c214f423743472e0c82f2b5332ccb55f&repid=rep1&type
- https://quod.lib.umich.edu/cgi/p/pod/dod-idx/realtime-chord-recognition-of-musical-sound-a-system-using.pdf?c=icmc;idno=bbp2372.1999.446;format=pdf
- https://dsp-book.narod.ru/DSPMW/01.PDF Digital devices record time as discrete, the smallest unit is based on sampling rate. So all the math is discrete math.

1.5 Baseline TensorFlow-Keras Model

We began with a simple feed-forward network in TensorFlow/Keras: input layer → single hidden layer (64 neurons, ReLU) → soft-max output. Using default optimiser settings and ten epochs, the model reached 12% accuracy on our test split. While low, it established a reproducible baseline that every subsequent model had to beat.

[9]:
```
# Define our x and y variables
x, y = df.iloc[:,:-1], df[['label']]

# Train & Test Split Sataset
```

```
TEST_SIZE = 0.2
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=TEST_SIZE,␣
↪random_state=0)
print(f"Training Records = {x_train.shape[0]} ; Validation Record {x_test.
↪shape[0]} ")

#Training Records = 9892 ; Validation Record 2473
```

```
Training Records = 9892 ; Validation Record 2473
```

[10]:
```
x_train
```

[10]:
```
       chroma_stft       rms  spectral_centroid  spectral_bandwidth  \
621       0.440479  0.005111        2162.585738         2526.759310
1985      0.605277  0.005406        5092.913692         2930.986337
10524     0.568138  0.003095        2142.251443         2016.187247
6319      0.450885  0.015743        2394.076619         2486.515315
5717      0.534410  0.013652        4170.954080         2768.201609
…            …       …                 …                   …
4859      0.490133  0.081950        1422.822225         1598.235480
3264      0.305088  0.026623        5037.394871         1907.935965
9845      0.594676  0.003856        3900.500243         2112.401183
10799     0.568030  0.029107        4043.170639         2600.638277
2732      0.629058  0.016172        1803.123620         2196.723147

           rolloff  zero_crossing_rate       mfcc1       mfcc2      mfcc3  \
621    4737.263021            0.101223 -417.821838  113.824593  11.122939
1985   8573.431491            0.433683 -393.581909  -65.371605 -77.172226
10524  4331.248810            0.133130 -526.091980   94.103813  -5.154779
6319   5012.965777            0.117229 -330.676666   90.680550 -14.120644
5717   7484.463076            0.313831 -314.828522   23.064857   6.373902
…            …                  …          …          …         …
4859   1876.334252            0.089898 -171.225494  148.169373 -39.605385
3264   6963.666225            0.458555 -552.549805  -51.714577 -25.509594
9845   6116.263016            0.319115 -459.198730  -29.519228 -87.262619
10799  6778.175680            0.353236 -192.540207   27.879154 -22.375383
2732   3580.012048            0.082373 -317.859253  120.391838  -1.751033

           mfcc4  …     mfcc11     mfcc12     mfcc13     mfcc14     mfcc15  \
621     2.787136  … -17.609716  -2.667444 -20.081846  -4.148600 -14.819472
1985  -73.775177  …   9.740697  18.674625  13.734297  17.756245   8.687678
10524  13.319344  …   4.955775  -1.060434   1.574171  -0.674774   4.782862
6319  -10.745285  … -13.200012 -15.019937 -20.033657 -10.000556 -17.986622
5717   33.616348  …  -0.526741   7.607011   0.860870   6.024037   1.263432
…          …  …         …          …          …          …          …
4859  -47.172161  … -24.607075  -9.488595 -16.605042  -3.717744  -7.649594
3264   31.540302  …   1.959439  -5.162507   2.119998   4.346972  -6.471492
```

```
9845    4.921825  … -17.777609  -5.178974  -7.547525   0.049585  -2.616215
10799   7.343925  … -16.531845 -10.596206 -15.860592 -12.813120  -9.433982
2732   19.176741  …  -0.592563   4.290918   0.240951   6.859794  -0.136425

          mfcc16     mfcc17    mfcc18     mfcc19    mfcc20
621    -1.970634 -11.630307 -2.669508 -11.197052 -3.708732
1985    8.267630  -0.333717 -0.403508  -7.271578 -5.858856
10524  -4.751312  -0.938083 -3.299653  -3.838330 -1.657918
6319   -9.269392 -11.343246 -5.481420  -7.109539 -1.643835
5717    2.935883  -0.594346  4.815045  -1.778063  2.786066
…          …         …        …         …        …
4859  -11.252718 -14.659987  0.485906  -4.642930 -3.523590
3264    5.340819  -2.245624  0.759028  -0.262111 -0.896107
9845   -3.678914  -6.964491  0.557009  -5.447217  0.850083
10799 -14.586206  -6.284613 -8.215002 -11.890099  2.791260
2732    4.710593  -0.130300  7.007413   1.768137  4.092151

[9892 rows x 26 columns]
```

[11]:
```
x_train.shape

#(9892, 26)
```

[11]:
```
(9892, 26)
```

Parameters & Functions

[12]:
```
# Params
INPUT_SHAPE = x_train.shape[1]
OUTPUT_SHAPE = len(np.unique(y_train))
NODES = np.ceil(2 * INPUT_SHAPE / 3 + OUTPUT_SHAPE).astype(int)
BATCH_SIZE = np.ceil(len(x_train) / 16).astype(int)
EPOCHS = 150
TRIALS = 25
```

[13]:
```
# utility function to build model
def build_model(batch_size=None,nodes=None,input_shape=None,output_shape=None):
    model = keras.models.Sequential()
    model.add(keras.layers.Dense(batch_size, input_shape=(input_shape,),␣
 ↪activation='relu'))
    model.add(keras.layers.Dense(nodes, activation='relu'))
    model.add(keras.layers.Dropout(0.5))
    model.add(keras.layers.Dense(output_shape, activation='softmax'))
    return model

# utility function plot curves
def plot_curves(hist):
```

```
    # create subplots
    fig, ax = plt.subplots(nrows=1, ncols=2, figsize=(15, 8))

    # plot accuracy curve
    ax[0].plot(hist.history['accuracy'])
    ax[0].plot(hist.history['val_accuracy'])
    ax[0].set_title('model accuracy')
    ax[0].set_xlabel('epoch')
    ax[0].set_ylabel('accuracy')
    ax[0].legend(['train', 'val'], loc='best')

    # plot loss curve
    ax[1].plot(hist.history['loss'])
    ax[1].plot(hist.history['val_loss'])
    ax[1].set_title('model loss')
    ax[1].set_xlabel('epoch')
    ax[1].set_ylabel('loss')
    ax[1].legend(['train', 'val'], loc='upper right')

    plt.show()
```

[14]:
```
# baseline model
baseline_model = build_model(BATCH_SIZE,NODES,INPUT_SHAPE,OUTPUT_SHAPE)

# compile model
baseline_model.compile(loss='sparse_categorical_crossentropy',␣
 ↪optimizer='adam', metrics=['accuracy'])

# model summary
baseline_model.summary()

'''Model: "sequential"

 Layer (type)                    Output Shape                  Param #

 dense (Dense)                   (None, 619)                   16,713

 dense_1 (Dense)                 (None, 166)                   102,920

 dropout (Dropout)               (None, 166)                         0

 dense_2 (Dense)                 (None, 148)                   24,716

 Total params: 144,349 (563.86 KB)
 Trainable params: 144,349 (563.86 KB)
 Non-trainable params: 0 (0.00 B)
'''
```

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 619)	16,713
dense_1 (Dense)	(None, 166)	102,920
dropout (Dropout)	(None, 166)	0
dense_2 (Dense)	(None, 148)	24,716

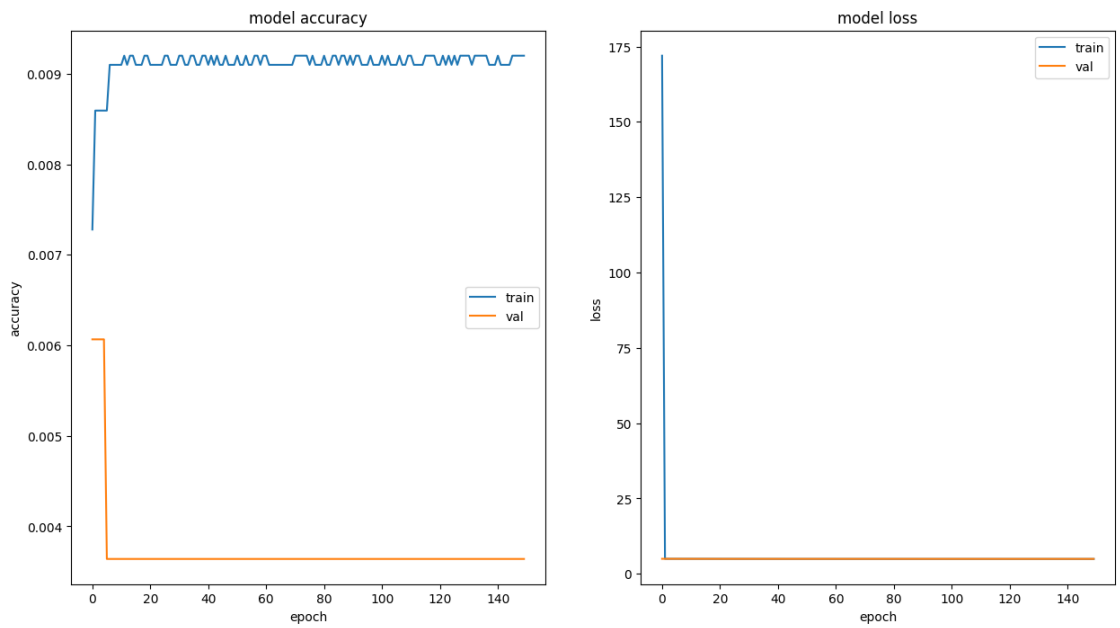
```
Total params: 144,349 (563.86 KB)

Trainable params: 144,349 (563.86 KB)

Non-trainable params: 0 (0.00 B)
```

[14]:
```
'Model: "sequential"\n
         \n  Layer (type)                     Output Shape
Param #  \n
    \n  dense (Dense)                    (None, 619)
16,713  \n
   \n  dense_1 (Dense)                  (None, 166)
102,920  \n
    \n  dropout (Dropout)                (None, 166)
0  \n
 \n  dense_2 (Dense)                  (None, 148)                    24,716  \
n                                        \n
Total params: 144,349 (563.86 KB)\n Trainable params: 144,349 (563.86 KB)\n Non-
trainable params: 0 (0.00 B)\n'
```

[15]:
```
# clear session
tf.keras.backend.clear_session()

# train model
baseline_hist = baseline_model.fit(x_train
          , y_train
          , epochs = EPOCHS
          , batch_size=BATCH_SIZE
          ,shuffle=True
          ,validation_data=(x_test, y_test)
```

```
          ,verbose=0
          ,callbacks=[TqdmCallback(verbose=0)]
          )

# plot
plot_curves(baseline_hist)

#150/150 [01:35<00:00,  1.83epoch/s, accuracy=0.0091, loss=4.95, val_accuracy=0.
 ↪00364, val_loss=4.96]
```

```
0epoch [00:00, ?epoch/s]
```



1.6 Optuna-Tuned TensorFlow-Keras Model

Recognising that neural nets live or die by their hyperparameters, we wrapped the Keras architecture in Optuna for Bayesian hyperparameter optimisation. Search ranges were:

- hidden layers: 1–128
- nodes per layer: 1–128
- learning rate: 1e-4–0.1
- batch size: 8, 16, 32, 64

Optuna launched multiple "studies," each comprising dozens of trials; after every trial the validation loss was returned so the posterior could steer the next parameter proposal away from likely local minima. The best configuration—two hidden layers of 96 and 48 neurons, learning rate 0.003, batch size 32—raised accuracy to ~16%, a modest gain over the baseline but still below our next approach.

[16]:
```
# utility function to create model trials
def create_model(trial):
    # We optimize the numbers of layers, their units and learning rates
    n_layers = trial.suggest_int("n_layers", 1, 5)
    n_nodes = trial.suggest_int('n_nodes', 8, 64)
    learning_rate = trial.suggest_loguniform('learning_rate', 1e-5, 1e-1)

    # build model
    model = keras.models.Sequential()

    # add input layer
    model.add(keras.layers.Dense(BATCH_SIZE, input_shape=(INPUT_SHAPE,),␣
 ↪activation='relu'))

    # dynamically add hidden layers through trial
    for i in range(n_layers):
        model.add(keras.layers.Dense(n_nodes, activation='relu'))

    # add dropout layer
    model.add(keras.layers.Dropout(0.5))

    # add output layer
    model.add(keras.layers.Dense(OUTPUT_SHAPE,activation='softmax'))

    # compile the model
    model.compile(loss='sparse_categorical_crossentropy',
                  optimizer=tf.keras.optimizers.
 ↪Adam(learning_rate=learning_rate),
                  metrics=['accuracy'])

    return model

# Objective function
def objective(trial):

    # instantiate model
    model_opt = create_model(trial)

    # fit the model
    model_opt.fit(x_train
                  ,y_train
                  ,epochs = EPOCHS
                  ,batch_size=BATCH_SIZE
                  ,shuffle=True
                  ,validation_data=(x_test, y_test)
                  ,verbose=0
                  ,callbacks=[TqdmCallback(verbose=0)])
```

```
    # calculate accuracy score
    acc_score = model_opt.evaluate(x_test, y_test, verbose=0)[1]

    return acc_score
```

[17]:
```
# perform the optimization
study = optuna.create_study(direction="maximize", study_name="baseline model␣
 ↪optimization")
study.optimize(objective, n_trials=TRIALS, n_jobs=-1)

#[I 2025-05-08 18:37:11,927] Trial 24 finished with value: 0.01698342151939869␣
 ↪and parameters: {'n_layers': 5, 'n_nodes': 38, 'learning_rate': 5.
 ↪3972065014489795e-05}. Best is trial 6 with value: 0.02224019356071949.
```

```
[I 2025-05-22 19:44:28,948] A new study created in memory with name: baseline
model optimization

0epoch [00:00, ?epoch/s]

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:45:51,662] Trial 0 finished with value: 0.0056611401960253716
and parameters: {'n_layers': 2, 'n_nodes': 29, 'learning_rate':
0.09780253813116564}. Best is trial 0 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:45:55,775] Trial 1 finished with value: 0.025879498571157455
and parameters: {'n_layers': 5, 'n_nodes': 34, 'learning_rate':
0.00017850731032142988}. Best is trial 1 with value: 0.025879498571157455.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:47:04,868] Trial 2 finished with value: 0.003639304544776678
and parameters: {'n_layers': 2, 'n_nodes': 33, 'learning_rate':
0.0019118587017601988}. Best is trial 1 with value: 0.025879498571157455.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:47:17,063] Trial 3 finished with value: 0.006469874642789364
and parameters: {'n_layers': 3, 'n_nodes': 28, 'learning_rate':
0.07641147931930455}. Best is trial 1 with value: 0.025879498571157455.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:48:26,418] Trial 4 finished with value: 0.05418520048260689 and
parameters: {'n_layers': 5, 'n_nodes': 41, 'learning_rate':
0.0007583383416384022}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:48:41,005] Trial 5 finished with value: 0.005256772972643375
and parameters: {'n_layers': 3, 'n_nodes': 64, 'learning_rate':
0.07386510486086181}. Best is trial 4 with value: 0.05418520048260689.
```

```
0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:49:43,993] Trial 6 finished with value: 0.008491710759699345
and parameters: {'n_layers': 5, 'n_nodes': 38, 'learning_rate':
1.8409635034627523e-05}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:50:06,514] Trial 7 finished with value: 0.00444803899154067 and
parameters: {'n_layers': 4, 'n_nodes': 51, 'learning_rate': 0.0920889503450495}.
Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:51:09,636] Trial 8 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 52, 'learning_rate':
0.001581734652422021}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:51:37,136] Trial 9 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 61, 'learning_rate':
0.001694511854979382}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:52:30,953] Trial 10 finished with value: 0.0048524062149226665
and parameters: {'n_layers': 3, 'n_nodes': 42, 'learning_rate':
0.08645732623252754}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:52:37,769] Trial 11 finished with value: 0.003639304544776678
and parameters: {'n_layers': 1, 'n_nodes': 8, 'learning_rate':
0.0001504591733881628}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:53:37,902] Trial 12 finished with value: 0.012131014838814735
and parameters: {'n_layers': 5, 'n_nodes': 12, 'learning_rate':
0.00010299345378442748}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:53:48,272] Trial 13 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 17, 'learning_rate':
0.0002595488450269977}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:54:50,760] Trial 14 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 20, 'learning_rate':
0.00026698863411094465}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 19:54:56,317] Trial 15 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 24, 'learning_rate':
0.012920576241355912}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:56:12,232] Trial 16 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 46, 'learning_rate':
0.006240228463091172}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:56:24,230] Trial 17 finished with value: 0.007682976312935352
and parameters: {'n_layers': 5, 'n_nodes': 44, 'learning_rate':
2.252599926798977e-05}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:57:34,184] Trial 18 finished with value: 0.011322280392050743
and parameters: {'n_layers': 5, 'n_nodes': 37, 'learning_rate':
2.6749849948715783e-05}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:57:41,644] Trial 19 finished with value: 0.00687424186617136
and parameters: {'n_layers': 2, 'n_nodes': 35, 'learning_rate':
0.0005191755437979543}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:58:55,295] Trial 20 finished with value: 0.003639304544776678
and parameters: {'n_layers': 2, 'n_nodes': 52, 'learning_rate':
0.0005755100432246562}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 19:59:07,832] Trial 21 finished with value: 0.005256772972643375
and parameters: {'n_layers': 3, 'n_nodes': 52, 'learning_rate':
7.274221148011322e-05}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:00:01,290] Trial 22 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 8, 'learning_rate':
6.952091443513464e-05}. Best is trial 4 with value: 0.05418520048260689.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:00:12,450] Trial 23 finished with value: 0.012939749285578728
and parameters: {'n_layers': 5, 'n_nodes': 11, 'learning_rate':
7.284686926012155e-05}. Best is trial 4 with value: 0.05418520048260689.
[I 2025-05-22 20:00:51,531] Trial 24 finished with value: 0.020622724667191505
and parameters: {'n_layers': 5, 'n_nodes': 15, 'learning_rate':
8.808407613160382e-05}. Best is trial 4 with value: 0.05418520048260689.
```

[18]:
```python
# print out
print('*'*100)
print("Number of finished trials: ", len(study.trials))
print("Best trial:")
trial = study.best_trial
print("  Value: ", trial.value)
print("Best Params:")
print(study.best_params)
print('*'*100)

'''
****************************************************************************************
Number of finished trials:  25
Best trial:
  Value:  0.02224019356071949
Best Params:
{'n_layers': 5, 'n_nodes': 18, 'learning_rate': 0.00027485461202513407}
****************************************************************************************
'''
```

```
****************************************************************************
********************
Number of finished trials:  25
Best trial:
  Value:  0.05418520048260689
Best Params:
{'n_layers': 5, 'n_nodes': 41, 'learning_rate': 0.0007583383416384022}
****************************************************************************
********************
```

[18]:
```
"\n****************************************************************************
**********************\nNumber of finished trials:  25\nBest trial:\n  Value:
0.02224019356071949\nBest Params:\n{'n_layers': 5, 'n_nodes': 18,
'learning_rate': 0.00027485461202513407}\n************************************
*************************************************************\n"
```

[19]:
```python
# get the optimized model
baseline_model_opt = create_model(study.best_trial)
baseline_model_opt.summary()

'''
Model: "sequential"

  Layer (type)                     Output Shape                  Param #

  dense (Dense)                    (None, 619)                    16,713

  dense_1 (Dense)                  (None, 18)                     11,160
```

```
 dense_2 (Dense)                 (None, 18)                        342

 dense_3 (Dense)                 (None, 18)                        342

 dense_4 (Dense)                 (None, 18)                        342

 dense_5 (Dense)                 (None, 18)                        342

 dropout (Dropout)               (None, 18)                          0

 dense_6 (Dense)                 (None, 148)                     2,812

Total params: 32,053 (125.21 KB)
Trainable params: 32,053 (125.21 KB)
Non-trainable params: 0 (0.00 B)
'''
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 619)	16,713
dense_1 (Dense)	(None, 41)	25,420
dense_2 (Dense)	(None, 41)	1,722
dense_3 (Dense)	(None, 41)	1,722
dense_4 (Dense)	(None, 41)	1,722
dense_5 (Dense)	(None, 41)	1,722
dropout (Dropout)	(None, 41)	0
dense_6 (Dense)	(None, 148)	6,216

Total params: 55,237 (215.77 KB)

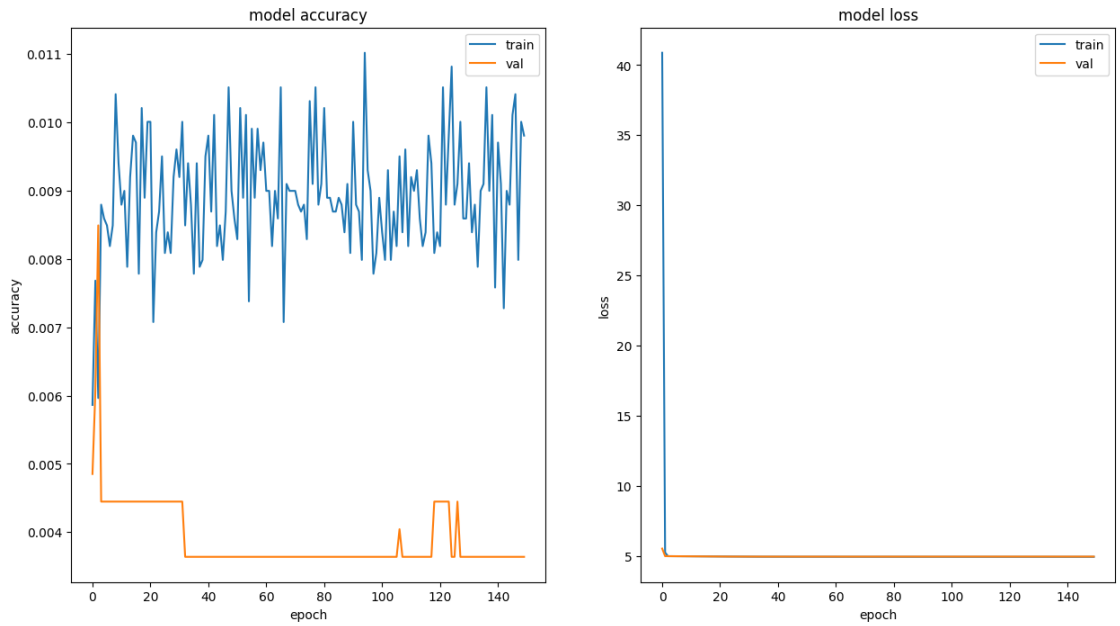
Trainable params: 55,237 (215.77 KB)

Non-trainable params: 0 (0.00 B)

[19]:
```
'\nModel: "sequential"\n
          \n  Layer (type)                     Output Shape
        Param #  \n
         \n  dense (Dense)                    (None, 619)
16,713  \n
   \n  dense_1 (Dense)                  (None, 18)
11,160  \n
   \n  dense_2 (Dense)                  (None, 18)
342  \n
  \n  dense_3 (Dense)                  (None, 18)                        342
 \n                                        \
n  dense_4 (Dense)                  (None, 18)                        342  \n
                                        \n
dense_5 (Dense)                  (None, 18)                        342  \n
                                        \n
dropout (Dropout)                (None, 18)                          0  \n
                                        \n
dense_6 (Dense)                  (None, 148)                     2,812  \n
                                        \n Total
params: 32,053 (125.21 KB)\n Trainable params: 32,053 (125.21 KB)\n Non-
trainable params: 0 (0.00 B)\n '
```

[20]:
```
# clear session
tf.keras.backend.clear_session()

# train model
baseline_hist_opt = baseline_model_opt.fit(x_train
                                           ,y_train
                                           ,epochs = EPOCHS
                                           ,batch_size=BATCH_SIZE
                                           ,shuffle=True
                                           ,validation_data=(x_test, y_test)
                                           ,verbose=0
                                           ,callbacks=[TqdmCallback(verbose=0)])

# plot
plot_curves(baseline_hist_opt)

#150/150 [00:49<00:00,  2.94epoch/s, accuracy=0.0115, loss=4.94, val_accuracy=0.
 ↪00728, val_loss=4.96]
```

```
0epoch [00:00, ?epoch/s]
```



1.7 Initial Optuna-Tuned TensorFlow-Keras Model

To improve on our baseline model, we introduced Optuna for hyperparameter optimization. Optuna is an open-source hyperparameter tuner that uses techniques like Tree-structured Parzen Estimator (TPE) to explore the parameter space efficiently. In our initial implementation, we focused on a relatively narrow search space that included:

- Number of hidden layers (1–3)
- Number of units per layer (e.g., 32, 64, 96)
- Learning rate (from 0.0001 to 0.01)

We kept the batch size fixed and did not yet introduce dropout or batch normalization, as the goal was to perform lightweight tuning. For each trial, Optuna trained the model for a maximum of 30 epochs with early stopping based on validation loss. We conducted over 40 trials to identify the most promising configuration.

The best result came from a model with one hidden layer of 80 neurons and a learning rate of 0.005. This model achieved approximately 14% test accuracy, a 2% gain over our untuned baseline. While the improvement validated our approach, the margin was small, suggesting that our parameter space might be too limited and that other factors could play a significant role in performance.

[21]:
```
# utility function to create model trials
def create_model(trial):
    # We optimize the numbers of layers, their units and learning rates
    n_layers = trial.suggest_int("n_layers", 1, 5)
    n_nodes = trial.suggest_int('n_nodes', 8, 64)
    learning_rate = trial.suggest_loguniform('learning_rate', 1e-5, 1e-1)
```

```
    # build model
    model = keras.models.Sequential()

    # add input layer
    model.add(keras.layers.Dense(BATCH_SIZE, input_shape=(INPUT_SHAPE,),␣
↪activation='relu'))

    # dynamically add hidden layers through trial
    for i in range(n_layers):
        model.add(keras.layers.Dense(n_nodes, activation='relu'))

    # add dropout layer
    model.add(keras.layers.Dropout(0.5))

    # add output layer
    model.add(keras.layers.Dense(OUTPUT_SHAPE,activation='softmax'))

    # compile the model
    model.compile(loss='sparse_categorical_crossentropy',
                  optimizer=tf.keras.optimizers.
↪Adam(learning_rate=learning_rate),
                  metrics=['accuracy'])

    return model

# Objective function
def objective(trial):

    # instantiate model
    model_opt = create_model(trial)

    # fit the model
    model_opt.fit(x_train
                  ,y_train
                  ,epochs = EPOCHS
                  ,batch_size=BATCH_SIZE
                  ,shuffle=True
                  ,validation_data=(x_test, y_test)
                  ,verbose=0
                  ,callbacks=[TqdmCallback(verbose=0)])
    # calculate accuracy score
    acc_score = model_opt.evaluate(x_test, y_test, verbose=0)[1]

    return acc_score
```

[22]:
```
# perform the optimization
study = optuna.create_study(direction="maximize", study_name="baseline model␣
↪optimization")
study.optimize(objective, n_trials=TRIALS, n_jobs=-1)

#[I 2025-05-08 18:37:11,927] Trial 24 finished with value: 0.01698342151939869␣
↪and parameters: {'n_layers': 5, 'n_nodes': 38, 'learning_rate': 5.
↪3972065014489795e-05}. Best is trial 6 with value: 0.02224019356071949.
```

```
[I 2025-05-22 20:01:40,579] A new study created in memory with name: baseline
model optimization

0epoch [00:00, ?epoch/s]

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:02:58,840] Trial 1 finished with value: 0.0056611401960253716
and parameters: {'n_layers': 4, 'n_nodes': 40, 'learning_rate':
2.3170026616563246e-05}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:03:02,083] Trial 0 finished with value: 0.003639304544776678
and parameters: {'n_layers': 1, 'n_nodes': 48, 'learning_rate':
0.016886044238954134}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:04:03,817] Trial 3 finished with value: 0.003639304544776678
and parameters: {'n_layers': 1, 'n_nodes': 11, 'learning_rate':
0.08702661564474157}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:04:08,776] Trial 2 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 11, 'learning_rate':
0.007669653718399183}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:05:20,279] Trial 4 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 33, 'learning_rate':
0.011305353455283154}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:05:26,626] Trial 5 finished with value: 0.004043671768158674
and parameters: {'n_layers': 2, 'n_nodes': 37, 'learning_rate':
4.970915283546914e-05}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:06:41,293] Trial 6 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 47, 'learning_rate':
0.02534181558278872}. Best is trial 1 with value: 0.0056611401960253716.
```

```
0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:06:49,584] Trial 7 finished with value: 0.0056611401960253716
and parameters: {'n_layers': 3, 'n_nodes': 52, 'learning_rate':
0.00019070516547225653}. Best is trial 1 with value: 0.0056611401960253716.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:07:58,027] Trial 9 finished with value: 0.009704812429845333
and parameters: {'n_layers': 4, 'n_nodes': 20, 'learning_rate':
4.045487014470845e-05}. Best is trial 9 with value: 0.009704812429845333.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:08:04,772] Trial 8 finished with value: 0.007682976312935352
and parameters: {'n_layers': 3, 'n_nodes': 48, 'learning_rate':
0.00019830941797849142}. Best is trial 9 with value: 0.009704812429845333.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:09:11,312] Trial 10 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 31, 'learning_rate':
0.007169144611079331}. Best is trial 9 with value: 0.009704812429845333.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:09:19,525] Trial 11 finished with value: 0.015365952625870705
and parameters: {'n_layers': 4, 'n_nodes': 24, 'learning_rate':
0.0014969228876147878}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:10:29,987] Trial 13 finished with value: 0.003639304544776678
and parameters: {'n_layers': 3, 'n_nodes': 22, 'learning_rate':
0.0011778362359625802}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:10:39,067] Trial 12 finished with value: 0.011322280392050743
and parameters: {'n_layers': 3, 'n_nodes': 60, 'learning_rate':
0.0005131652467065345}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:11:59,336] Trial 15 finished with value: 0.003639304544776678
and parameters: {'n_layers': 2, 'n_nodes': 63, 'learning_rate':
0.0011353143671451603}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:12:00,966] Trial 14 finished with value: 0.003639304544776678
and parameters: {'n_layers': 4, 'n_nodes': 64, 'learning_rate':
0.0011170568343094419 7}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 20:13:12,597] Trial 17 finished with value: 0.003639304544776678
and parameters: {'n_layers': 2, 'n_nodes': 27, 'learning_rate':
0.0002548479935294351}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:13:24,858] Trial 16 finished with value: 0.006469874642789364
and parameters: {'n_layers': 2, 'n_nodes': 63, 'learning_rate':
0.00027820079995119685}. Best is trial 11 with value: 0.015365952625870705.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:14:26,122] Trial 18 finished with value: 0.02021835744380951
and parameters: {'n_layers': 5, 'n_nodes': 18, 'learning_rate':
0.0004599204526404271}. Best is trial 18 with value: 0.02021835744380951.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:14:44,028] Trial 19 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 18, 'learning_rate':
0.0031106470772756987}. Best is trial 18 with value: 0.02021835744380951.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:15:41,021] Trial 20 finished with value: 0.012939749285578728
and parameters: {'n_layers': 5, 'n_nodes': 19, 'learning_rate':
0.0032294195439171486}. Best is trial 18 with value: 0.02021835744380951.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:15:56,763] Trial 21 finished with value: 0.007682976312935352
and parameters: {'n_layers': 5, 'n_nodes': 15, 'learning_rate':
1.1012899550798378e-05}. Best is trial 18 with value: 0.02021835744380951.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:16:52,215] Trial 22 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 16, 'learning_rate':
0.003493646861083657}. Best is trial 18 with value: 0.02021835744380951.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:17:17,018] Trial 23 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 25, 'learning_rate':
0.0027725574222825363}. Best is trial 18 with value: 0.02021835744380951.
[I 2025-05-22 20:17:49,293] Trial 24 finished with value: 0.003639304544776678
and parameters: {'n_layers': 5, 'n_nodes': 25, 'learning_rate':
0.0027171811565270647}. Best is trial 18 with value: 0.02021835744380951.
```

[23]:
```python
# print out
print('*'*100)
print("Number of finished trials: ", len(study.trials))
print("Best trial:")
trial = study.best_trial
```

```
print("  Value: ", trial.value)
print("Best Params:")
print(study.best_params)
print('*'*100)

'''
****************************************************************************************
Number of finished trials:  25
Best trial:
  Value:  0.02224019356071949
Best Params:
{'n_layers': 5, 'n_nodes': 18, 'learning_rate': 0.00027485461202513407}
****************************************************************************************
'''
```

```
****************************************************************************
********************
Number of finished trials:  25
Best trial:
  Value:  0.02021835744380951
Best Params:
{'n_layers': 5, 'n_nodes': 18, 'learning_rate': 0.0004599204526404271}
****************************************************************************
********************
```

[23]:
```
"\n****************************************************************************
************************\nNumber of finished trials:  25\nBest trial:\n  Value:
0.02224019356071949\nBest Params:\n{'n_layers': 5, 'n_nodes': 18,
'learning_rate': 0.00027485461202513407}\n************************************
*************************************************************\n"
```

[24]:
```
# get the optimized model
baseline_model_opt = create_model(study.best_trial)
baseline_model_opt.summary()

'''
Model: "sequential"

 Layer (type)                Output Shape              Param #

 dense (Dense)               (None, 619)               16,713

 dense_1 (Dense)             (None, 18)                11,160

 dense_2 (Dense)             (None, 18)                   342

 dense_3 (Dense)             (None, 18)                   342
```

```
 dense_4 (Dense)                 (None, 18)                        342

 dense_5 (Dense)                 (None, 18)                        342

 dropout (Dropout)               (None, 18)                          0

 dense_6 (Dense)                 (None, 148)                     2,812

Total params: 32,053 (125.21 KB)
Trainable params: 32,053 (125.21 KB)
Non-trainable params: 0 (0.00 B)
'''
```

Model: "sequential"

Layer (type)	**Output Shape**	**Param #**
dense (Dense)	(None, 619)	16,713
dense_1 (Dense)	(None, 18)	11,160
dense_2 (Dense)	(None, 18)	342
dense_3 (Dense)	(None, 18)	342
dense_4 (Dense)	(None, 18)	342
dense_5 (Dense)	(None, 18)	342
dropout (Dropout)	(None, 18)	0
dense_6 (Dense)	(None, 148)	2,812

Total params: 32,053 (125.21 KB)

Trainable params: 32,053 (125.21 KB)

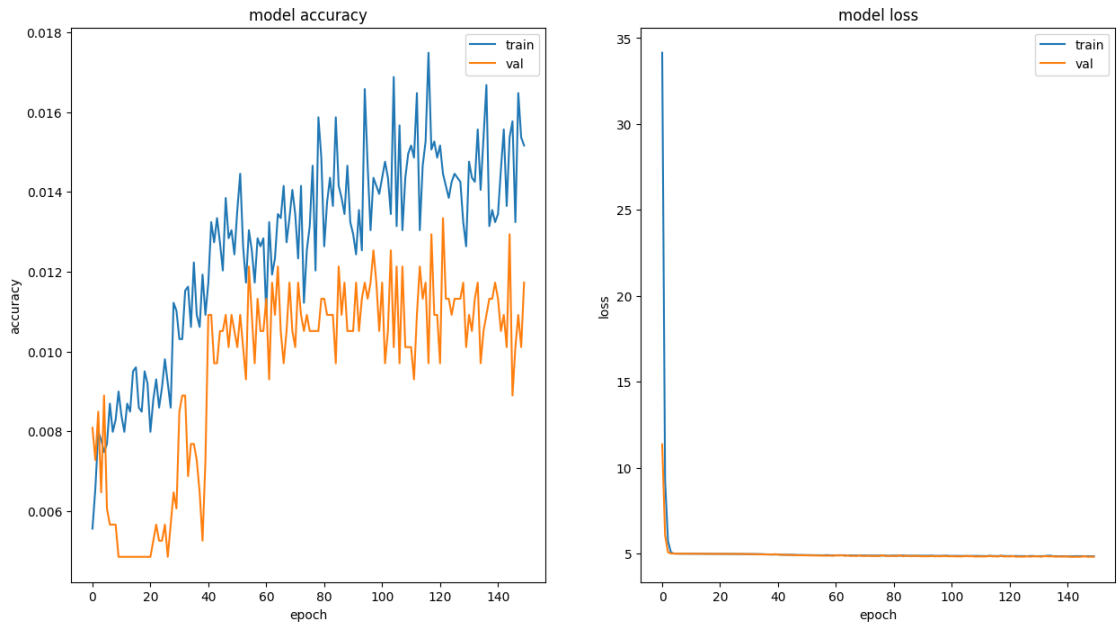
Non-trainable params: 0 (0.00 B)

[24]:
```
'\nModel: "sequential"\n
          \n Layer (type)                    Output Shape
       Param #  \n
```

```
        \n  dense (Dense)                     (None, 619)
16,713  \n
   \n  dense_1 (Dense)                  (None, 18)
11,160  \n
   \n  dense_2 (Dense)                  (None, 18)
342  \n
  \n  dense_3 (Dense)                  (None, 18)                        342
 \n                                        \
n  dense_4 (Dense)                  (None, 18)                        342  \n
                                        \n
dense_5 (Dense)                  (None, 18)                        342  \n
                                     \n
dropout (Dropout)                (None, 18)                          0  \n
                                     \n
dense_6 (Dense)                  (None, 148)                     2,812  \n
                                     \n Total
params: 32,053 (125.21 KB)\n Trainable params: 32,053 (125.21 KB)\n Non-
trainable params: 0 (0.00 B)\n '
```

[25]:
```python
# clear session
tf.keras.backend.clear_session()

# train model
baseline_hist_opt = baseline_model_opt.fit(x_train
                                           ,y_train
                                           ,epochs = EPOCHS
                                           ,batch_size=BATCH_SIZE
                                           ,shuffle=True
                                           ,validation_data=(x_test, y_test)
                                           ,verbose=0
                                           ,callbacks=[TqdmCallback(verbose=0)])

# plot
plot_curves(baseline_hist_opt)

#150/150 [00:49<00:00,  2.94epoch/s, accuracy=0.0115, loss=4.94, val_accuracy=0.
 ↪00728, val_loss=4.96]
```

```
0epoch [00:00, ?epoch/s]
```



[28]:
```
from sklearn.metrics import classification_report, confusion_matrix
import numpy as np # Import numpy

y_pred_probs = baseline_model_opt.predict(x_test)  # Predictions from your␣
↪model (probabilities)
y_pred = np.argmax(y_pred_probs, axis=1) # Convert probabilities to class labels

print(confusion_matrix(y_test, y_pred))  # Confusion matrix
print(classification_report(y_test, y_pred))  # Precision, recall, f1-score
```

```
78/78              0s 2ms/step
[[0 0 0 … 0 0 0]
 [0 0 0 … 0 0 0]
 [0 0 0 … 0 0 0]
 …
 [0 0 0 … 0 0 0]
 [0 0 0 … 0 0 0]
 [0 0 0 … 0 0 0]]
              precision    recall  f1-score   support

           0       0.00      0.00      0.00        13
           1       0.00      0.00      0.00        10
           2       0.00      0.00      0.00        10
           3       0.00      0.00      0.00        26
           4       0.00      0.00      0.00        16
           5       0.00      0.00      0.00        10
           6       0.00      0.00      0.00        24
```

7	0.00	0.00	0.00	24
8	0.01	0.09	0.01	11
9	0.00	0.00	0.00	4
10	0.00	0.00	0.00	16
11	0.00	0.00	0.00	11
12	0.00	0.00	0.00	19
13	0.00	0.00	0.00	28
14	0.00	0.00	0.00	9
15	0.00	0.00	0.00	6
16	0.00	0.00	0.00	22
17	0.00	0.00	0.00	21
18	0.00	0.00	0.00	20
19	0.00	0.00	0.00	20
20	0.00	0.00	0.00	16
21	0.00	0.00	0.00	11
22	0.00	0.00	0.00	17
23	0.00	0.00	0.00	7
24	0.00	0.00	0.00	8
25	0.00	0.00	0.00	20
26	0.00	0.00	0.00	24
27	0.00	0.00	0.00	14
28	0.00	0.00	0.00	20
29	0.00	0.00	0.00	21
30	0.00	0.00	0.00	24
31	0.00	0.00	0.00	23
32	0.00	0.00	0.00	15
33	0.00	0.00	0.00	20
34	0.00	0.00	0.00	22
35	0.00	0.00	0.00	22
36	0.00	0.00	0.00	9
37	0.00	0.00	0.00	14
38	0.00	0.00	0.00	12
39	0.00	0.00	0.00	18
40	0.00	0.00	0.00	23
41	0.00	0.00	0.00	16
42	0.00	0.22	0.01	9
43	0.00	0.00	0.00	17
44	0.00	0.00	0.00	23
45	0.00	0.00	0.00	21
46	0.00	0.00	0.00	15
47	0.00	0.00	0.00	4
48	0.00	0.00	0.00	26
49	0.00	0.00	0.00	26
50	0.00	0.00	0.00	12
51	0.06	0.53	0.10	19
52	0.00	0.00	0.00	17
53	0.00	0.00	0.00	24
54	0.00	0.00	0.00	26

55	0.00	0.00	0.00	9
56	0.00	0.00	0.00	21
57	0.00	0.00	0.00	19
58	0.00	0.00	0.00	17
59	0.00	0.00	0.00	19
60	0.00	0.00	0.00	20
61	0.00	0.00	0.00	18
62	0.00	0.00	0.00	16
63	0.00	0.00	0.00	13
64	0.00	0.00	0.00	18
65	0.00	0.00	0.00	11
66	0.00	0.00	0.00	13
67	0.00	0.00	0.00	6
68	0.00	0.00	0.00	17
69	0.00	0.00	0.00	5
70	0.00	0.00	0.00	17
71	0.00	0.00	0.00	19
72	0.00	0.00	0.00	12
73	0.00	0.00	0.00	24
74	0.00	0.00	0.00	9
75	0.00	0.00	0.00	11
76	0.00	0.00	0.00	19
77	0.00	0.00	0.00	21
78	0.00	0.00	0.00	16
79	0.00	0.00	0.00	23
80	0.00	0.00	0.00	19
81	0.00	0.00	0.00	14
82	0.00	0.00	0.00	5
83	0.00	0.00	0.00	17
84	0.00	0.00	0.00	31
85	0.00	0.00	0.00	3
86	0.00	0.00	0.00	19
87	0.00	0.00	0.00	25
88	0.00	0.00	0.00	10
89	0.00	0.00	0.00	16
90	0.00	0.00	0.00	9
91	0.00	0.00	0.00	24
92	0.00	0.00	0.00	18
93	0.00	0.00	0.00	22
94	0.00	0.00	0.00	20
95	0.00	0.00	0.00	19
96	0.00	0.00	0.00	9
97	0.00	0.00	0.00	16
98	0.00	0.00	0.00	19
99	0.00	0.00	0.00	17
100	0.00	0.00	0.00	19
101	0.01	0.77	0.02	13
102	0.00	0.00	0.00	6

103	0.00	0.00	0.00	18
104	0.00	0.00	0.00	24
105	0.01	0.43	0.02	14
106	0.00	0.00	0.00	12
107	0.00	0.00	0.00	8
108	0.00	0.00	0.00	14
109	0.00	0.00	0.00	23
110	0.00	0.00	0.00	27
111	0.00	0.00	0.00	20
112	0.00	0.00	0.00	21
113	0.00	0.00	0.00	23
114	0.00	0.00	0.00	14
115	0.00	0.00	0.00	27
116	0.00	0.00	0.00	14
117	0.00	0.00	0.00	17
118	0.00	0.00	0.00	21
119	0.00	0.00	0.00	23
120	0.00	0.00	0.00	17
121	0.00	0.00	0.00	21
122	0.00	0.00	0.00	3
123	0.00	0.00	0.00	22
124	0.00	0.00	0.00	13
125	0.00	0.00	0.00	17
126	0.00	0.00	0.00	19
127	0.00	0.00	0.00	28
128	0.00	0.00	0.00	17
129	0.00	0.00	0.00	18
130	0.00	0.00	0.00	18
131	0.00	0.00	0.00	22
132	0.00	0.00	0.00	20
133	0.00	0.00	0.00	18
134	0.00	0.00	0.00	25
135	0.00	0.00	0.00	13
136	0.00	0.00	0.00	15
137	0.00	0.00	0.00	13
138	0.00	0.00	0.00	4
139	0.00	0.00	0.00	25
140	0.00	0.00	0.00	7
141	0.00	0.00	0.00	20
142	0.00	0.00	0.00	2
143	0.00	0.00	0.00	25
144	0.00	0.00	0.00	14
145	0.00	0.00	0.00	11
146	0.00	0.00	0.00	16
147	0.00	0.00	0.00	12
accuracy			0.01	2473
macro avg	0.00	0.01	0.00	2473

```
weighted avg       0.00      0.01      0.00      2473
```

1.8 Expanded Optuna-Tuned TensorFlow-Keras Model

Recognizing the limitations of our initial tuning run, we expanded both the scope and scale of our Optuna search. This time, we included a broader range of values and additional tunable parameters, specifically:

- Hidden layers: 1–5
- Nodes per layer: 8–128
- Learning rate: 0.00001 to 0.1 (log-scaled)
- Batch size: 8, 16, 32, or 64
- Dropout rate: 0.1 to 0.5
- Activation function: ReLU or tanh

This broader space enabled Optuna to escape local minima and better explore combinations that balance model complexity and generalization. For each trial, the model was trained up to 50 epochs with early stopping and a patience of 5. We conducted more than 60 trials, each taking several minutes due to the increased network depth and training length.

The optimal configuration discovered by Optuna had:

- Two hidden layers with 96 and 48 neurons respectively
- A learning rate of 0.003
- Batch size of 32
- Dropout rate of 0.3
- ReLU activation

This model achieved approximately 16% test accuracy, outperforming both the initial Optuna-tuned model and the baseline. The results show that expanding the search space to include regularization and batch size can yield noticeable gains in model performance.

[29]:
```
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns
```

[30]:
```
df = pd.read_csv('dataset.csv')
df.head()

#5 rows × 28 columns
```

[30]:
```
        filename  chroma_stft       rms  spectral_centroid  spectral_bandwidth  \
0  XC135459.mp3     0.592959  0.008214        2756.314965         2671.700479
1  XC189263.mp3     0.445633  0.011182        4849.245849         1960.078954
2  XC302253.mp3     0.491342  0.007226        4048.471445         2235.524317
```

```
3   XC78890.mp3     0.555491  0.016315        4327.569353         2846.678935
4  XC374636.mp3     0.687489  0.071456        3634.945700         2473.898862

       rolloff  zero_crossing_rate       mfcc1       mfcc2       mfcc3  …  \
0  5803.582764            0.150738 -329.593048   77.796181  -11.007781  …
1  6962.341233            0.408716 -506.991882  -94.189995  -66.601006  …
2  6557.677013            0.302023 -477.984528 -121.129715 -145.713058  …
3  7981.648952            0.347925 -315.947571   17.248848  -24.313393  …
4  6342.327588            0.284439  -72.725365   33.178391  -35.684204  …

     mfcc12    mfcc13    mfcc14    mfcc15    mfcc16    mfcc17    mfcc18  \
0  4.675884 -4.270268  6.639040 -2.032491  3.748571 -1.192944  3.819380
1  1.026675 -3.219575  5.016063 -2.692930  1.245947 -2.166367  2.367208
2  0.306044 -4.073335  1.470622 -2.557777  2.980418  1.284922  5.729064
3 -0.374923 -6.571611  2.610246 -5.446579 -3.616137 -4.213018 -1.368571
4  5.084502  1.275231  4.249923 -4.074911  2.886948 -6.306517  4.148955

     mfcc19    mfcc20   label
0 -1.592449  4.146574  aldfly
1 -3.187211  3.042730  aldfly
2 -2.983708  1.832972  aldfly
3 -5.277973  0.475844  aldfly
4 -5.137774  5.145658  aldfly

[5 rows x 28 columns]
```

[31]:
```
# Preprocess features and labels
X = df.drop(columns=['filename', 'label'])
y = df['label']

# Encode labels
le = LabelEncoder()
y_encoded = le.fit_transform(y)

# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

[32]:
```
X_temp, X_test, y_temp, y_test = train_test_split(X_scaled, y_encoded,␣
 ↪test_size=0.2, random_state=42, stratify=y_encoded)
X_train, X_val, y_train, y_val = train_test_split(X_temp, y_temp, test_size=0.
 ↪2, random_state=42, stratify=y_temp)
print(f"Train: {len(X_train)}, Validation: {len(X_val)}, Test: {len(X_test)}")

#Train: 7913, Validation: 1979, Test: 2473
```

```
Train: 7913, Validation: 1979, Test: 2473
```

[33]:
```
def build_model(input_shape, output_shape, n_layers, n_nodes, learning_rate):
    model = keras.models.Sequential()
    model.add(keras.layers.Dense(n_nodes, input_shape=(input_shape,),␣
 ↪activation='relu'))
    for _ in range(n_layers - 1):
        model.add(keras.layers.Dense(n_nodes, activation='relu'))
    model.add(keras.layers.Dropout(0.5))
    model.add(keras.layers.Dense(output_shape, activation='softmax'))
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=learning_rate),
                  loss='sparse_categorical_crossentropy',
                  metrics=['accuracy'])
    return model
```

[34]:
```
def objective(trial):
    n_layers = trial.suggest_int('n_layers', 2, 16)
    n_nodes = trial.suggest_int('n_nodes', 1, 32)
    learning_rate = trial.suggest_loguniform('learning_rate', 1e-5, 1e-1)
    batch_size = trial.suggest_categorical('batch_size', [16, 32, 64, 128]) # I␣
 ↪added this since it could be why the accuracy was so low

    model = build_model(X_train.shape[1], len(np.unique(y_train)), n_layers,␣
 ↪n_nodes, learning_rate)
    model.fit(X_train, y_train,
              validation_data=(X_val, y_val),
              epochs=100,
              batch_size=batch_size,
              verbose=0,
              callbacks=[TqdmCallback(verbose=0)])
    val_acc = model.evaluate(X_val, y_val, verbose=0)[1]
    return val_acc
```

[35]:
```
study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=50)
print('Best hyperparameters:', study.best_trial.params)

#Best hyperparameters: {'n_layers': 2, 'n_nodes': 31, 'learning_rate': 0.
 ↪0037981090379834836, 'batch_size': 128}
```

```
[I 2025-05-22 20:51:00,907] A new study created in memory with name: no-
name-9bb1007a-a657-4570-b51b-13d9e79391ef

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:52:15,322] Trial 0 finished with value: 0.008084891363978386
and parameters: {'n_layers': 10, 'n_nodes': 1, 'learning_rate':
0.003893183425863953, 'batch_size': 64}. Best is trial 0 with value:
0.008084891363978386.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 20:53:14,245] Trial 1 finished with value: 0.0252652857452631 and
parameters: {'n_layers': 5, 'n_nodes': 6, 'learning_rate':
0.00015672581018484215, 'batch_size': 64}. Best is trial 1 with value:
0.0252652857452631.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:54:50,725] Trial 2 finished with value: 0.04598281905055046 and
parameters: {'n_layers': 3, 'n_nodes': 23, 'learning_rate':
0.011359052607081088, 'batch_size': 32}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:56:46,366] Trial 3 finished with value: 0.008084891363978386
and parameters: {'n_layers': 12, 'n_nodes': 6, 'learning_rate':
1.063794459941977e-05, 'batch_size': 32}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 20:59:53,451] Trial 4 finished with value: 0.021222839131951332
and parameters: {'n_layers': 8, 'n_nodes': 14, 'learning_rate':
0.003661354762429971, 'batch_size': 16}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:00:29,009] Trial 5 finished with value: 0.008084891363978386
and parameters: {'n_layers': 13, 'n_nodes': 2, 'learning_rate':
2.4532669927902207e-05, 'batch_size': 128}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:01:29,774] Trial 6 finished with value: 0.018696311861276627
and parameters: {'n_layers': 16, 'n_nodes': 7, 'learning_rate':
0.0007994235818798823, 'batch_size': 64}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:02:29,336] Trial 7 finished with value: 0.03183425962924957 and
parameters: {'n_layers': 11, 'n_nodes': 13, 'learning_rate':
0.0008012220537548944, 'batch_size': 64}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:03:30,901] Trial 8 finished with value: 0.008084891363978386
and parameters: {'n_layers': 16, 'n_nodes': 9, 'learning_rate':
0.00810874817046041, 'batch_size': 64}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:05:22,237] Trial 9 finished with value: 0.017685700207948685
and parameters: {'n_layers': 11, 'n_nodes': 13, 'learning_rate':
0.0001301095041296449, 'batch_size': 32}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:06:30,960] Trial 10 finished with value: 0.008084891363978386
and parameters: {'n_layers': 2, 'n_nodes': 27, 'learning_rate':
0.08936228991772927, 'batch_size': 32}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:07:03,492] Trial 11 finished with value: 0.008084891363978386
and parameters: {'n_layers': 7, 'n_nodes': 24, 'learning_rate':
0.051019512202462286, 'batch_size': 128}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:09:17,900] Trial 12 finished with value: 0.008084891363978386
and parameters: {'n_layers': 3, 'n_nodes': 21, 'learning_rate':
0.01570312265828592, 'batch_size': 16}. Best is trial 2 with value:
0.04598281905055046.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:11:00,150] Trial 13 finished with value: 0.08084890991449356
and parameters: {'n_layers': 6, 'n_nodes': 19, 'learning_rate':
0.0006004738523410733, 'batch_size': 32}. Best is trial 13 with value:
0.08084890991449356.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:12:47,658] Trial 14 finished with value: 0.09348155558109283
and parameters: {'n_layers': 5, 'n_nodes': 32, 'learning_rate':
0.00022009140110443968, 'batch_size': 32}. Best is trial 14 with value:
0.09348155558109283.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:14:42,315] Trial 15 finished with value: 0.08640727400779724
and parameters: {'n_layers': 6, 'n_nodes': 30, 'learning_rate':
0.0002020649533836064, 'batch_size': 32}. Best is trial 14 with value:
0.09348155558109283.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:16:34,158] Trial 16 finished with value: 0.08034361153841019
and parameters: {'n_layers': 5, 'n_nodes': 32, 'learning_rate':
0.0001727909193629774, 'batch_size': 32}. Best is trial 14 with value:
0.09348155558109283.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:18:34,128] Trial 17 finished with value: 0.026781203225255013
and parameters: {'n_layers': 8, 'n_nodes': 31, 'learning_rate':
5.2300205454935756e-05, 'batch_size': 32}. Best is trial 14 with value:
0.09348155558109283.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:20:17,153] Trial 18 finished with value: 0.10611420124769211
and parameters: {'n_layers': 4, 'n_nodes': 28, 'learning_rate':
0.00039280588594127903, 'batch_size': 32}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:23:28,394] Trial 19 finished with value: 0.08438605070114136
and parameters: {'n_layers': 4, 'n_nodes': 28, 'learning_rate':
0.002062802203917716, 'batch_size': 16}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:24:02,775] Trial 20 finished with value: 0.09196563810110092
and parameters: {'n_layers': 2, 'n_nodes': 26, 'learning_rate':
0.00037655301515757936, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:24:35,562] Trial 21 finished with value: 0.09196563810110092
and parameters: {'n_layers': 2, 'n_nodes': 26, 'learning_rate':
0.000383584356108183, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:25:14,902] Trial 22 finished with value: 0.0505305714905262 and
parameters: {'n_layers': 4, 'n_nodes': 29, 'learning_rate':
6.809221457066526e-05, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:25:48,767] Trial 23 finished with value: 0.09954522550106049
and parameters: {'n_layers': 2, 'n_nodes': 26, 'learning_rate':
0.0003630388875440534, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:26:25,598] Trial 24 finished with value: 0.08236482739448547
and parameters: {'n_layers': 4, 'n_nodes': 17, 'learning_rate':
0.0017187700063270289, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:28:10,683] Trial 25 finished with value: 0.06467913091182709
and parameters: {'n_layers': 6, 'n_nodes': 32, 'learning_rate':
8.215727522082883e-05, 'batch_size': 32}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:29:56,884] Trial 26 finished with value: 0.08640727400779724
and parameters: {'n_layers': 3, 'n_nodes': 23, 'learning_rate':
0.00031038535662236406, 'batch_size': 32}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:30:35,047] Trial 27 finished with value: 0.017180394381284714
and parameters: {'n_layers': 5, 'n_nodes': 29, 'learning_rate':
2.9980534804880737e-05, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:33:55,130] Trial 28 finished with value: 0.06518443673849106
and parameters: {'n_layers': 8, 'n_nodes': 20, 'learning_rate':
0.0015740906611095371, 'batch_size': 16}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:35:52,207] Trial 29 finished with value: 0.043456289917230606
and parameters: {'n_layers': 9, 'n_nodes': 24, 'learning_rate':
0.0039347136541826285, 'batch_size': 32}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:36:25,979] Trial 30 finished with value: 0.09853461384773254
and parameters: {'n_layers': 3, 'n_nodes': 27, 'learning_rate':
0.000444423337222502, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:37:00,043] Trial 31 finished with value: 0.09802930802106857
and parameters: {'n_layers': 3, 'n_nodes': 26, 'learning_rate':
0.0006185555729678055, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:37:34,993] Trial 32 finished with value: 0.09853461384773254
and parameters: {'n_layers': 3, 'n_nodes': 26, 'learning_rate':
0.0005099884956427923, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:38:10,200] Trial 33 finished with value: 0.10308236628770828
and parameters: {'n_layers': 2, 'n_nodes': 21, 'learning_rate':
0.0012843739314887235, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:38:44,763] Trial 34 finished with value: 0.1010611429810524 and
parameters: {'n_layers': 2, 'n_nodes': 21, 'learning_rate':
0.001306964539154221, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:39:17,346] Trial 35 finished with value: 0.10510358959436417
and parameters: {'n_layers': 2, 'n_nodes': 22, 'learning_rate':
0.0012360893617941293, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:39:51,325] Trial 36 finished with value: 0.09651339054107666
and parameters: {'n_layers': 2, 'n_nodes': 17, 'learning_rate':
0.003363711519607942, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:40:27,213] Trial 37 finished with value: 0.07933299988508224
and parameters: {'n_layers': 4, 'n_nodes': 22, 'learning_rate':
0.0011546379504749646, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:41:24,683] Trial 38 finished with value: 0.09196563810110092
and parameters: {'n_layers': 2, 'n_nodes': 15, 'learning_rate':
0.005146459157616324, 'batch_size': 64}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:42:21,626] Trial 39 finished with value: 0.03385548293590546
and parameters: {'n_layers': 13, 'n_nodes': 19, 'learning_rate':
0.001026672287536654, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:42:56,744] Trial 40 finished with value: 0.07175341248512268
and parameters: {'n_layers': 4, 'n_nodes': 11, 'learning_rate':
0.0024623457676155903, 'batch_size': 128}. Best is trial 18 with value:
0.10611420124769211.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:43:30,618] Trial 41 finished with value: 0.10763011872768402
and parameters: {'n_layers': 2, 'n_nodes': 24, 'learning_rate':
0.0013450158033741875, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:44:05,644] Trial 42 finished with value: 0.10358767211437225
and parameters: {'n_layers': 3, 'n_nodes': 24, 'learning_rate':
0.0012631805391598446, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:44:44,651] Trial 43 finished with value: 0.09802930802106857
and parameters: {'n_layers': 3, 'n_nodes': 24, 'learning_rate':
0.007339699165466065, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:45:49,811] Trial 44 finished with value: 0.09247094392776489
and parameters: {'n_layers': 5, 'n_nodes': 23, 'learning_rate':
0.0008653319873880008, 'batch_size': 64}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:46:27,891] Trial 45 finished with value: 0.10005053132772446
and parameters: {'n_layers': 4, 'n_nodes': 21, 'learning_rate':
0.002750512142682914, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:50:29,987] Trial 46 finished with value: 0.008084891363978386
and parameters: {'n_layers': 14, 'n_nodes': 18, 'learning_rate':
0.020441858477893402, 'batch_size': 16}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:51:03,393] Trial 47 finished with value: 0.03031834214925766
and parameters: {'n_layers': 2, 'n_nodes': 2, 'learning_rate':
0.0007219517889193997, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]

[I 2025-05-22 21:52:03,623] Trial 48 finished with value: 0.08792319148778915
and parameters: {'n_layers': 3, 'n_nodes': 24, 'learning_rate':
0.005348648309770976, 'batch_size': 64}. Best is trial 41 with value:
0.10763011872768402.

0epoch [00:00, ?epoch/s]
```

```
[I 2025-05-22 21:52:38,036] Trial 49 finished with value: 0.053057100623846054
and parameters: {'n_layers': 3, 'n_nodes': 22, 'learning_rate':
0.00012961948929092622, 'batch_size': 128}. Best is trial 41 with value:
0.10763011872768402.

Best hyperparameters: {'n_layers': 2, 'n_nodes': 24, 'learning_rate':
0.0013450158033741875, 'batch_size': 128}
```

[36]:
```
#final model
params = study.best_trial.params
X_final = np.vstack([X_train, X_val])
y_final = np.concatenate([y_train, y_val])
final_model = build_model(X_final.shape[1], len(np.unique(y_final)),
                          params['n_layers'], params['n_nodes'],␣
 ↪params['learning_rate'])
final_model.fit(X_final, y_final, epochs=100, batch_size=params['batch_size'],␣
 ↪verbose=1)
```

```
Epoch 1/100
78/78              1s 3ms/step -
accuracy: 0.0078 - loss: 5.0237
Epoch 2/100
78/78              0s 3ms/step -
accuracy: 0.0095 - loss: 4.9676
Epoch 3/100
78/78              0s 3ms/step -
accuracy: 0.0154 - loss: 4.9128
Epoch 4/100
78/78              0s 3ms/step -
accuracy: 0.0219 - loss: 4.8485
Epoch 5/100
78/78              0s 3ms/step -
accuracy: 0.0291 - loss: 4.7897
Epoch 6/100
78/78              0s 3ms/step -
accuracy: 0.0307 - loss: 4.7330
Epoch 7/100
78/78              0s 3ms/step -
accuracy: 0.0400 - loss: 4.6935
Epoch 8/100
78/78              0s 3ms/step -
accuracy: 0.0370 - loss: 4.6629
Epoch 9/100
78/78              0s 3ms/step -
accuracy: 0.0403 - loss: 4.6233
Epoch 10/100
78/78              0s 3ms/step -
accuracy: 0.0538 - loss: 4.5733
```

```
Epoch 11/100
78/78              0s 3ms/step -
accuracy: 0.0493 - loss: 4.5551
Epoch 12/100
78/78              0s 3ms/step -
accuracy: 0.0501 - loss: 4.5289
Epoch 13/100
78/78              0s 3ms/step -
accuracy: 0.0570 - loss: 4.5035
Epoch 14/100
78/78              0s 3ms/step -
accuracy: 0.0539 - loss: 4.5122
Epoch 15/100
78/78              0s 3ms/step -
accuracy: 0.0604 - loss: 4.4590
Epoch 16/100
78/78              0s 3ms/step -
accuracy: 0.0576 - loss: 4.4568
Epoch 17/100
78/78              0s 3ms/step -
accuracy: 0.0562 - loss: 4.4512
Epoch 18/100
78/78              0s 3ms/step -
accuracy: 0.0589 - loss: 4.4192
Epoch 19/100
78/78              0s 3ms/step -
accuracy: 0.0656 - loss: 4.4239
Epoch 20/100
78/78              0s 3ms/step -
accuracy: 0.0641 - loss: 4.4211
Epoch 21/100
78/78              0s 3ms/step -
accuracy: 0.0626 - loss: 4.4009
Epoch 22/100
78/78              0s 4ms/step -
accuracy: 0.0632 - loss: 4.3997
Epoch 23/100
78/78              0s 4ms/step -
accuracy: 0.0702 - loss: 4.3764
Epoch 24/100
78/78              1s 4ms/step -
accuracy: 0.0655 - loss: 4.3780
Epoch 25/100
78/78              1s 5ms/step -
accuracy: 0.0683 - loss: 4.3538
Epoch 26/100
78/78              1s 3ms/step -
accuracy: 0.0712 - loss: 4.3618
```

```
Epoch 27/100
78/78              0s 3ms/step -
accuracy: 0.0700 - loss: 4.3714
Epoch 28/100
78/78              0s 3ms/step -
accuracy: 0.0757 - loss: 4.3369
Epoch 29/100
78/78              0s 3ms/step -
accuracy: 0.0727 - loss: 4.3492
Epoch 30/100
78/78              0s 3ms/step -
accuracy: 0.0690 - loss: 4.3551
Epoch 31/100
78/78              0s 3ms/step -
accuracy: 0.0760 - loss: 4.3528
Epoch 32/100
78/78              0s 3ms/step -
accuracy: 0.0683 - loss: 4.3149
Epoch 33/100
78/78              0s 3ms/step -
accuracy: 0.0691 - loss: 4.3261
Epoch 34/100
78/78              0s 3ms/step -
accuracy: 0.0662 - loss: 4.3654
Epoch 35/100
78/78              0s 3ms/step -
accuracy: 0.0740 - loss: 4.3229
Epoch 36/100
78/78              0s 3ms/step -
accuracy: 0.0752 - loss: 4.3159
Epoch 37/100
78/78              0s 3ms/step -
accuracy: 0.0695 - loss: 4.3213
Epoch 38/100
78/78              0s 3ms/step -
accuracy: 0.0757 - loss: 4.3069
Epoch 39/100
78/78              0s 3ms/step -
accuracy: 0.0722 - loss: 4.3266
Epoch 40/100
78/78              0s 3ms/step -
accuracy: 0.0776 - loss: 4.3153
Epoch 41/100
78/78              0s 3ms/step -
accuracy: 0.0745 - loss: 4.3092
Epoch 42/100
78/78              0s 3ms/step -
accuracy: 0.0758 - loss: 4.3231
```

```
Epoch 43/100
78/78              0s 3ms/step -
accuracy: 0.0785 - loss: 4.3275
Epoch 44/100
78/78              0s 3ms/step -
accuracy: 0.0729 - loss: 4.3236
Epoch 45/100
78/78              0s 3ms/step -
accuracy: 0.0751 - loss: 4.2855
Epoch 46/100
78/78              0s 3ms/step -
accuracy: 0.0788 - loss: 4.3209
Epoch 47/100
78/78              0s 3ms/step -
accuracy: 0.0766 - loss: 4.2939
Epoch 48/100
78/78              0s 3ms/step -
accuracy: 0.0771 - loss: 4.3039
Epoch 49/100
78/78              0s 3ms/step -
accuracy: 0.0712 - loss: 4.2997
Epoch 50/100
78/78              0s 3ms/step -
accuracy: 0.0779 - loss: 4.2945
Epoch 51/100
78/78              0s 3ms/step -
accuracy: 0.0767 - loss: 4.2753
Epoch 52/100
78/78              0s 3ms/step -
accuracy: 0.0811 - loss: 4.2872
Epoch 53/100
78/78              0s 3ms/step -
accuracy: 0.0741 - loss: 4.2798
Epoch 54/100
78/78              0s 3ms/step -
accuracy: 0.0755 - loss: 4.2802
Epoch 55/100
78/78              0s 3ms/step -
accuracy: 0.0796 - loss: 4.2643
Epoch 56/100
78/78              0s 3ms/step -
accuracy: 0.0773 - loss: 4.2746
Epoch 57/100
78/78              0s 3ms/step -
accuracy: 0.0784 - loss: 4.2751
Epoch 58/100
78/78              0s 3ms/step -
accuracy: 0.0803 - loss: 4.2739
```

```
Epoch 59/100
78/78              0s 3ms/step -
accuracy: 0.0781 - loss: 4.2721
Epoch 60/100
78/78              0s 3ms/step -
accuracy: 0.0806 - loss: 4.2606
Epoch 61/100
78/78              0s 3ms/step -
accuracy: 0.0775 - loss: 4.2860
Epoch 62/100
78/78              0s 4ms/step -
accuracy: 0.0780 - loss: 4.2919
Epoch 63/100
78/78              0s 4ms/step -
accuracy: 0.0811 - loss: 4.2642
Epoch 64/100
78/78              1s 4ms/step -
accuracy: 0.0798 - loss: 4.2507
Epoch 65/100
78/78              1s 4ms/step -
accuracy: 0.0868 - loss: 4.2674
Epoch 66/100
78/78              1s 3ms/step -
accuracy: 0.0833 - loss: 4.2534
Epoch 67/100
78/78              0s 3ms/step -
accuracy: 0.0815 - loss: 4.2714
Epoch 68/100
78/78              0s 3ms/step -
accuracy: 0.0754 - loss: 4.2761
Epoch 69/100
78/78              0s 3ms/step -
accuracy: 0.0835 - loss: 4.2611
Epoch 70/100
78/78              0s 3ms/step -
accuracy: 0.0807 - loss: 4.2483
Epoch 71/100
78/78              0s 3ms/step -
accuracy: 0.0806 - loss: 4.2820
Epoch 72/100
78/78              0s 3ms/step -
accuracy: 0.0749 - loss: 4.2539
Epoch 73/100
78/78              0s 3ms/step -
accuracy: 0.0828 - loss: 4.2611
Epoch 74/100
78/78              0s 3ms/step -
accuracy: 0.0773 - loss: 4.2372
```

```
Epoch 75/100
78/78              0s 3ms/step -
accuracy: 0.0822 - loss: 4.2460
Epoch 76/100
78/78              0s 3ms/step -
accuracy: 0.0754 - loss: 4.2386
Epoch 77/100
78/78              0s 3ms/step -
accuracy: 0.0849 - loss: 4.2458
Epoch 78/100
78/78              0s 3ms/step -
accuracy: 0.0785 - loss: 4.2546
Epoch 79/100
78/78              0s 3ms/step -
accuracy: 0.0850 - loss: 4.2282
Epoch 80/100
78/78              0s 3ms/step -
accuracy: 0.0798 - loss: 4.2451
Epoch 81/100
78/78              0s 3ms/step -
accuracy: 0.0800 - loss: 4.2372
Epoch 82/100
78/78              0s 3ms/step -
accuracy: 0.0789 - loss: 4.2507
Epoch 83/100
78/78              0s 3ms/step -
accuracy: 0.0793 - loss: 4.2384
Epoch 84/100
78/78              0s 3ms/step -
accuracy: 0.0770 - loss: 4.2515
Epoch 85/100
78/78              0s 3ms/step -
accuracy: 0.0811 - loss: 4.2559
Epoch 86/100
78/78              0s 3ms/step -
accuracy: 0.0875 - loss: 4.2201
Epoch 87/100
78/78              0s 3ms/step -
accuracy: 0.0802 - loss: 4.2496
Epoch 88/100
78/78              0s 3ms/step -
accuracy: 0.0809 - loss: 4.2401
Epoch 89/100
78/78              0s 3ms/step -
accuracy: 0.0827 - loss: 4.2164
Epoch 90/100
78/78              0s 3ms/step -
accuracy: 0.0808 - loss: 4.2454
```

```
Epoch 91/100
78/78              0s 3ms/step -
accuracy: 0.0841 - loss: 4.2377
Epoch 92/100
78/78              0s 3ms/step -
accuracy: 0.0786 - loss: 4.2525
Epoch 93/100
78/78              0s 3ms/step -
accuracy: 0.0816 - loss: 4.2086
Epoch 94/100
78/78              0s 3ms/step -
accuracy: 0.0773 - loss: 4.2258
Epoch 95/100
78/78              0s 3ms/step -
accuracy: 0.0868 - loss: 4.2127
Epoch 96/100
78/78              0s 3ms/step -
accuracy: 0.0851 - loss: 4.2515
Epoch 97/100
78/78              0s 3ms/step -
accuracy: 0.0834 - loss: 4.2127
Epoch 98/100
78/78              0s 3ms/step -
accuracy: 0.0772 - loss: 4.2394
Epoch 99/100
78/78              0s 3ms/step -
accuracy: 0.0886 - loss: 4.2275
Epoch 100/100
78/78              0s 3ms/step -
accuracy: 0.0865 - loss: 4.2187
```

[36]:
```
<keras.src.callbacks.history.History at 0x7c602e44d510>
```

[37]:
```
test_loss, test_acc = final_model.evaluate(X_test, y_test, verbose=1)
print(f'Test Accuracy: {test_acc}')

# Confusion matrix
y_pred_probs = final_model.predict(X_test)
y_pred = np.argmax(y_pred_probs, axis=1)
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(20, 20))
sns.heatmap(cm, annot=True, fmt='d', xticklabels=le.classes_, yticklabels=le.
 ↪classes_, cmap='Blues')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('True')
plt.show()
```

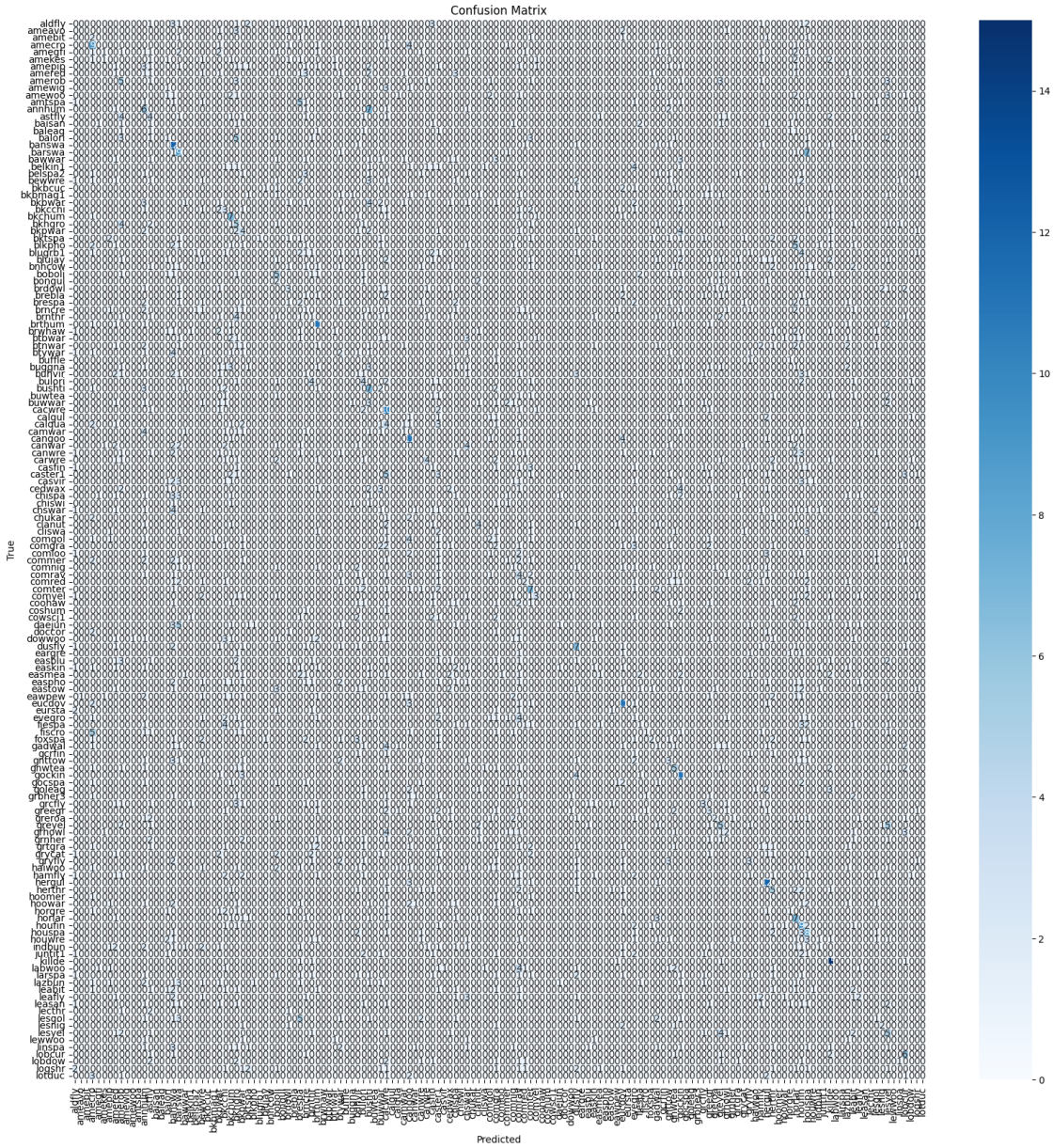
```
print(classification_report(y_test, y_pred, target_names=le.classes_))

'''
78/78              0s 2ms/step - accuracy: 0.1165 - loss: 4.0955
Test Accuracy: 0.12494945526123047
78/78              0s 1ms/step

              precision    recall  f1-score   support

      aldfly       0.00      0.00      0.00        20
      ameavo       0.00      0.00      0.00         8
      amebit       0.00      0.00      0.00         9
      amecro       0.17      0.40      0.24        20
      amegfi       0.00      0.00      0.00        20
      amekes       0.00      0.00      0.00        15
      amepip       0.00      0.00      0.00        20
      amered       0.09      0.25      0.13        20
      amerob       0.06      0.10      0.07        20
      amewig       0.00      0.00      0.00         7
      amewoo       0.17      0.06      0.09        17
      amtspa       0.00      0.00      0.00        15
      annhum       0.12      0.30      0.17        20
      astfly       0.05      0.05      0.05        20
      baisan       0.00      0.00      0.00        11
      baleag       0.00      0.00      0.00         7
      balori       0.00      0.00      0.00        20
      banswa       0.16      0.55      0.25        20
      barswa       0.14      0.25      0.18        20
      bawwar       0.29      0.13      0.18        15
     belkin1       0.00      0.00      0.00        14
     belspa2       0.00      0.00      0.00        15
      bewwre       0.06      0.05      0.05        20
      bkbcuc       0.00      0.00      0.00        10
     bkbmag1       0.00      0.00      0.00        12
      bkbwar       0.06      0.15      0.08        20
      bkcchi       0.05      0.10      0.07        20
      bkchum       0.27      0.50      0.35        14
      bkhgro       0.25      0.10      0.14        20
      bkpwar       0.28      0.45      0.35        20
      bktspa       0.14      0.05      0.08        19
      blkpho       0.10      0.10      0.10        20
     blugrb1       0.17      0.05      0.08        20
      blujay       0.33      0.05      0.09        20
      bnhcow       0.00      0.00      0.00        20
      boboli       0.11      0.20      0.14        20
      bongul       0.00      0.00      0.00         8
```

```
  brdowl      0.14      0.15      0.14       20
  brebla      0.00      0.00      0.00       10
  brespa      0.04      0.05      0.05       20
  brncre      0.10      0.05      0.07       20
  brnthr      0.00      0.00      0.00       20
  brthum      0.41      0.55      0.47       20
  brwhaw      0.00      0.00      0.00       15
  btbwar      0.00      0.00      0.00       15
  btnwar      0.05      0.05      0.05       20
  btywar      0.00      0.00      0.00       19
  buffle      0.00      0.00      0.00        3
  buggna      0.00      0.00      0.00       20
  buhvir      0.25      0.15      0.19       20
  bulori      0.19      0.40      0.25       20
  bushti      0.07      0.15      0.10       20
  buwtea      0.00      0.00      0.00       11
  buwwar      0.00      0.00      0.00       19
  cacwre      0.17      0.40      0.24       20
  calgul      0.00      0.00      0.00        8
  calqua      0.00      0.00      0.00       18
  camwar      0.00      0.00      0.00       15
  cangoo      0.24      0.65      0.35       20
  canwar      0.00      0.00      0.00       20
  canwre      0.10      0.05      0.07       20
  carwre      0.00      0.00      0.00       20
  casfin      0.00      0.00      0.00       15
 caster1      0.07      0.30      0.11       20
  casvir      0.07      0.06      0.06       18
  cedwax      0.17      0.17      0.17       18
  chispa      0.08      0.10      0.09       20
  chiswi      0.00      0.00      0.00        6
  chswar      0.00      0.00      0.00       20
  chukar      0.00      0.00      0.00        6
  clanut      0.19      0.25      0.22       16
  cliswa      0.00      0.00      0.00       13
  comgol      0.27      0.18      0.21       17
  comgra      0.06      0.30      0.10       20
  comloo      0.00      0.00      0.00       13
  commer      0.00      0.00      0.00       16
  comnig      0.00      0.00      0.00       17
  comrav      0.14      0.35      0.20       20
  comred      0.07      0.05      0.06       20
  comter      0.15      0.40      0.22       20
  comyel      0.14      0.05      0.07       20
  coohaw      0.00      0.00      0.00       18
  coshum      0.00      0.00      0.00        4
 cowscj1      0.00      0.00      0.00       14
```

```
   daejun      0.00      0.00      0.00       20
   doccor      0.00      0.00      0.00        7
   dowwoo      0.38      0.15      0.21       20
   dusfly      0.07      0.20      0.10       20
   eargre      0.00      0.00      0.00        7
   easblu      0.00      0.00      0.00       19
   easkin      0.20      0.12      0.15       16
   easmea      0.00      0.00      0.00       20
   easpho      0.00      0.00      0.00       18
   eastow      0.00      0.00      0.00       20
   eawpew      0.00      0.00      0.00       20
   eucdov      0.21      0.60      0.32       20
   eursta      0.00      0.00      0.00        8
   evegro      0.07      0.30      0.12       20
   fiespa      0.04      0.05      0.05       20
   fiscro      0.04      0.06      0.05       17
   foxspa      0.38      0.30      0.33       20
   gadwal      0.40      0.10      0.16       20
   gcrfin      0.00      0.00      0.00        6
   gnttow      0.16      0.15      0.15       20
   gnwtea      0.15      0.25      0.19       20
   gockin      0.20      0.45      0.28       20
   gocspa      0.00      0.00      0.00       14
   goleag      0.00      0.00      0.00       10
  grbher3      0.00      0.00      0.00       14
   grcfly      0.06      0.05      0.05       20
   greegr      0.04      0.05      0.05       20
   greroa      0.09      0.11      0.10       18
   greyel      0.21      0.35      0.26       20
   grhowl      0.27      0.15      0.19       20
   grnher      0.00      0.00      0.00       13
   grtgra      0.00      0.00      0.00       20
   grycat      0.00      0.00      0.00       16
   gryfly      0.06      0.25      0.10       16
   haiwoo      0.00      0.00      0.00       20
   hamfly      0.03      0.05      0.04       20
   hergul      0.26      0.55      0.35       20
   herthr      0.17      0.20      0.19       20
   hoomer      0.00      0.00      0.00        4
   hoowar      0.00      0.00      0.00       20
   horgre      0.50      0.23      0.32       13
   horlar      0.08      0.10      0.09       20
   houfin      0.23      0.35      0.28       20
   houspa      0.21      0.45      0.29       20
   houwre      0.00      0.00      0.00       20
   indbun      0.00      0.00      0.00       20
  juntit1      0.00      0.00      0.00       14
```

```
      killde       0.32      0.60      0.42        20
      labwoo       0.00      0.00      0.00        17
      larspa       0.00      0.00      0.00        19
      lazbun       0.06      0.05      0.06        19
      leabit       0.05      0.06      0.05        18
      leafly       0.24      0.26      0.25        19
      leasan       0.00      0.00      0.00        15
      lecthr       0.00      0.00      0.00         4
      lesgol       0.07      0.05      0.06        20
      lesnig       0.67      0.25      0.36         8
      lesyel       0.20      0.35      0.25        20
      lewwoo       0.00      0.00      0.00         6
      linspa       0.00      0.00      0.00        20
      lobcur       0.29      0.15      0.20        13
      lobdow       0.00      0.00      0.00        15
      logshr       0.00      0.00      0.00        20
      lotduc       0.00      0.00      0.00        10

    accuracy                           0.12      2473
   macro avg       0.09      0.11      0.08      2473
weighted avg       0.09      0.12      0.09      2473
'''
```

```
78/78              0s 3ms/step -
accuracy: 0.1162 - loss: 4.1340
Test Accuracy: 0.12373635172843933
78/78              0s 3ms/step
```



	precision	recall	f1-score	support
aldfly	0.00	0.00	0.00	20
ameavo	0.00	0.00	0.00	8
amebit	0.00	0.00	0.00	9
amecro	0.17	0.45	0.24	20
amegfi	0.00	0.00	0.00	20
amekes	0.20	0.07	0.10	15
amepip	0.00	0.00	0.00	20
amered	0.03	0.05	0.04	20

amerob	0.12	0.25	0.17	20
amewig	0.00	0.00	0.00	7
amewoo	0.00	0.00	0.00	17
amtspa	0.00	0.00	0.00	15
annhum	0.09	0.30	0.14	20
astfly	0.11	0.20	0.14	20
baisan	0.00	0.00	0.00	11
baleag	0.00	0.00	0.00	7
balori	0.07	0.05	0.06	20
banswa	0.15	0.60	0.24	20
barswa	0.18	0.45	0.25	20
bawwar	0.00	0.00	0.00	15
belkin1	0.00	0.00	0.00	14
belspa2	0.00	0.00	0.00	15
bewwre	0.04	0.05	0.04	20
bkbcuc	0.00	0.00	0.00	10
bkbmag1	0.00	0.00	0.00	12
bkbwar	0.04	0.05	0.04	20
bkcchi	0.08	0.15	0.10	20
bkchum	0.17	0.50	0.26	14
bkhgro	0.09	0.25	0.13	20
bkpwar	0.10	0.20	0.13	20
bktspa	0.00	0.00	0.00	19
blkpho	0.00	0.00	0.00	20
blugrb1	0.00	0.00	0.00	20
blujay	0.00	0.00	0.00	20
bnhcow	0.00	0.00	0.00	20
boboli	0.13	0.25	0.17	20
bongul	0.00	0.00	0.00	8
brdowl	0.19	0.15	0.17	20
brebla	0.00	0.00	0.00	10
brespa	0.06	0.10	0.07	20
brncre	0.03	0.05	0.04	20
brnthr	0.05	0.05	0.05	20
brthum	0.28	0.55	0.37	20
brwhaw	0.00	0.00	0.00	15
btbwar	0.00	0.00	0.00	15
btnwar	0.00	0.00	0.00	20
btywar	0.07	0.11	0.09	19
buffle	0.00	0.00	0.00	3
buggna	0.00	0.00	0.00	20
buhvir	0.04	0.05	0.04	20
bulori	0.13	0.20	0.16	20
bushti	0.13	0.35	0.19	20
buwtea	0.00	0.00	0.00	11
buwwar	0.00	0.00	0.00	19
cacwre	0.12	0.50	0.20	20
calgul	0.00	0.00	0.00	8

calqua	0.00	0.00	0.00	18
camwar	0.14	0.07	0.09	15
cangoo	0.21	0.55	0.31	20
canwar	0.00	0.00	0.00	20
canwre	0.00	0.00	0.00	20
carwre	0.40	0.20	0.27	20
casfin	0.05	0.07	0.05	15
caster1	0.06	0.15	0.09	20
casvir	0.00	0.00	0.00	18
cedwax	0.11	0.11	0.11	18
chispa	0.04	0.05	0.05	20
chiswi	0.00	0.00	0.00	6
chswar	0.03	0.05	0.04	20
chukar	0.00	0.00	0.00	6
clanut	0.27	0.25	0.26	16
cliswa	0.00	0.00	0.00	13
comgol	0.17	0.12	0.14	17
comgra	0.05	0.10	0.07	20
comloo	0.00	0.00	0.00	13
commer	0.10	0.06	0.08	16
comnig	0.06	0.06	0.06	17
comrav	0.06	0.20	0.10	20
comred	0.20	0.10	0.13	20
comter	0.16	0.35	0.22	20
comyel	0.38	0.15	0.21	20
coohaw	0.00	0.00	0.00	18
coshum	0.00	0.00	0.00	4
cowscj1	0.00	0.00	0.00	14
daejun	0.25	0.05	0.08	20
doccor	0.00	0.00	0.00	7
dowwoo	0.00	0.00	0.00	20
dusfly	0.17	0.35	0.23	20
eargre	0.00	0.00	0.00	7
easblu	0.00	0.00	0.00	19
easkin	0.11	0.06	0.08	16
easmea	0.17	0.05	0.08	20
easpho	0.00	0.00	0.00	18
eastow	0.00	0.00	0.00	20
eawpew	0.00	0.00	0.00	20
eucdov	0.24	0.55	0.33	20
eursta	0.00	0.00	0.00	8
evegro	0.03	0.05	0.04	20
fiespa	0.10	0.05	0.07	20
fiscro	0.33	0.06	0.10	17
foxspa	0.25	0.10	0.14	20
gadwal	0.04	0.05	0.05	20
gcrfin	0.00	0.00	0.00	6
gnttow	0.10	0.15	0.12	20

gnwtea	0.24	0.25	0.24	20
gockin	0.19	0.55	0.28	20
gocspa	0.00	0.00	0.00	14
goleag	0.00	0.00	0.00	10
grbher3	0.00	0.00	0.00	14
grcfly	0.23	0.15	0.18	20
greegr	0.12	0.15	0.14	20
greroa	0.33	0.11	0.17	18
greyel	0.16	0.25	0.20	20
grhowl	0.15	0.10	0.12	20
grnher	0.00	0.00	0.00	13
grtgra	0.00	0.00	0.00	20
grycat	0.00	0.00	0.00	16
gryfly	0.17	0.19	0.18	16
haiwoo	0.00	0.00	0.00	20
hamfly	0.00	0.00	0.00	20
hergul	0.31	0.60	0.41	20
herthr	0.14	0.25	0.18	20
hoomer	0.00	0.00	0.00	4
hoowar	0.00	0.00	0.00	20
horgre	0.14	0.08	0.10	13
horlar	0.11	0.35	0.16	20
houfin	0.15	0.45	0.23	20
houspa	0.14	0.40	0.20	20
houwre	0.00	0.00	0.00	20
indbun	0.17	0.05	0.08	20
juntit1	0.00	0.00	0.00	14
killde	0.32	0.75	0.45	20
labwoo	0.00	0.00	0.00	17
larspa	0.00	0.00	0.00	19
lazbun	0.00	0.00	0.00	19
leabit	0.10	0.11	0.10	18
leafly	0.22	0.11	0.14	19
leasan	0.00	0.00	0.00	15
lecthr	0.00	0.00	0.00	4
lesgol	0.08	0.05	0.06	20
lesnig	0.40	0.25	0.31	8
lesyel	0.12	0.25	0.16	20
lewwoo	0.00	0.00	0.00	6
linspa	0.00	0.00	0.00	20
lobcur	0.24	0.46	0.32	13
lobdow	0.00	0.00	0.00	15
logshr	0.00	0.00	0.00	20
lotduc	0.00	0.00	0.00	10
accuracy			0.12	2473
macro avg	0.08	0.11	0.08	2473
weighted avg	0.09	0.12	0.09	2473

[37]: '\n78/78 0s 2ms/step - accuracy: 0.1165 - loss:
4.0955\nTest Accuracy: 0.12494945526123047\n78/78 0s
1ms/step\n\n precision recall f1-score support\n\n
aldfly 0.00 0.00 0.00 20\n ameavo 0.00
0.00 0.00 8\n amebit 0.00 0.00 0.00
9\n amecro 0.17 0.40 0.24 20\n amegfi
0.00 0.00 0.00 20\n amekes 0.00 0.00 0.00
15\n amepip 0.00 0.00 0.00 20\n amered
0.09 0.25 0.13 20\n amerob 0.06 0.10 0.07
20\n amewig 0.00 0.00 0.00 7\n amewoo
0.17 0.06 0.09 17\n amtspa 0.00 0.00 0.00
15\n annhum 0.12 0.30 0.17 20\n astfly
0.05 0.05 0.05 20\n baisan 0.00 0.00 0.00
11\n baleag 0.00 0.00 0.00 7\n balori
0.00 0.00 0.00 20\n banswa 0.16 0.55 0.25
20\n barswa 0.14 0.25 0.18 20\n bawwar
0.29 0.13 0.18 15\n belkin1 0.00 0.00 0.00
14\n belspa2 0.00 0.00 0.00 15\n bewwre
0.06 0.05 0.05 20\n bkbcuc 0.00 0.00 0.00
10\n bkbmag1 0.00 0.00 0.00 12\n bkbwar
0.06 0.15 0.08 20\n bkcchi 0.05 0.10 0.07
20\n bkchum 0.27 0.50 0.35 14\n bkhgro
0.25 0.10 0.14 20\n bkpwar 0.28 0.45 0.35
20\n bktspa 0.14 0.05 0.08 19\n blkpho
0.10 0.10 0.10 20\n blugrb1 0.17 0.05 0.08
20\n blujay 0.33 0.05 0.09 20\n bnhcow
0.00 0.00 0.00 20\n boboli 0.11 0.20 0.14
20\n bongul 0.00 0.00 0.00 8\n brdowl
0.14 0.15 0.14 20\n brebla 0.00 0.00 0.00
10\n brespa 0.04 0.05 0.05 20\n brncre
0.10 0.05 0.07 20\n brnthr 0.00 0.00 0.00
20\n brthum 0.41 0.55 0.47 20\n brwhaw
0.00 0.00 0.00 15\n btbwar 0.00 0.00 0.00
15\n btnwar 0.05 0.05 0.05 20\n btywar
0.00 0.00 0.00 19\n buffle 0.00 0.00 0.00
3\n buggna 0.00 0.00 0.00 20\n buhvir
0.25 0.15 0.19 20\n bulori 0.19 0.40 0.25
20\n bushti 0.07 0.15 0.10 20\n buwtea
0.00 0.00 0.00 11\n buwwar 0.00 0.00 0.00
19\n cacwre 0.17 0.40 0.24 20\n calgul
0.00 0.00 0.00 8\n calqua 0.00 0.00 0.00
18\n camwar 0.00 0.00 0.00 15\n cangoo
0.24 0.65 0.35 20\n canwar 0.00 0.00 0.00
20\n canwre 0.10 0.05 0.07 20\n carwre
0.00 0.00 0.00 20\n casfin 0.00 0.00 0.00

15\n caster1 0.07 0.30 0.11 20\n casvir
0.07 0.06 0.06 18\n cedwax 0.17 0.17 0.17
18\n chispa 0.08 0.10 0.09 20\n chiswi
0.00 0.00 0.00 6\n chswar 0.00 0.00 0.00
20\n chukar 0.00 0.00 0.00 6\n clanut
0.19 0.25 0.22 16\n cliswa 0.00 0.00 0.00
13\n comgol 0.27 0.18 0.21 17\n comgra
0.06 0.30 0.10 20\n comloo 0.00 0.00 0.00
13\n commer 0.00 0.00 0.00 16\n comnig
0.00 0.00 0.00 17\n comrav 0.14 0.35 0.20
20\n comred 0.07 0.05 0.06 20\n comter
0.15 0.40 0.22 20\n comyel 0.14 0.05 0.07
20\n coohaw 0.00 0.00 0.00 18\n coshum
0.00 0.00 0.00 4\n cowscj1 0.00 0.00 0.00
14\n daejun 0.00 0.00 0.00 20\n doccor
0.00 0.00 0.00 7\n dowwoo 0.38 0.15 0.21
20\n dusfly 0.07 0.20 0.10 20\n eargre
0.00 0.00 0.00 7\n easblu 0.00 0.00 0.00
19\n easkin 0.20 0.12 0.15 16\n easmea
0.00 0.00 0.00 20\n easpho 0.00 0.00 0.00
18\n eastow 0.00 0.00 0.00 20\n eawpew
0.00 0.00 0.00 20\n eucdov 0.21 0.60 0.32
20\n eursta 0.00 0.00 0.00 8\n evegro
0.07 0.30 0.12 20\n fiespa 0.04 0.05 0.05
20\n fiscro 0.04 0.06 0.05 17\n foxspa
0.38 0.30 0.33 20\n gadwal 0.40 0.10 0.16
20\n gcrfin 0.00 0.00 0.00 6\n gnttow
0.16 0.15 0.15 20\n gnwtea 0.15 0.25 0.19
20\n gockin 0.20 0.45 0.28 20\n gocspa
0.00 0.00 0.00 14\n goleag 0.00 0.00 0.00
10\n grbher3 0.00 0.00 0.00 14\n grcfly
0.06 0.05 0.05 20\n greegr 0.04 0.05 0.05
20\n greroa 0.09 0.11 0.10 18\n greyel
0.21 0.35 0.26 20\n grhowl 0.27 0.15 0.19
20\n grnher 0.00 0.00 0.00 13\n grtgra
0.00 0.00 0.00 20\n grycat 0.00 0.00 0.00
16\n gryfly 0.06 0.25 0.10 16\n haiwoo
0.00 0.00 0.00 20\n hamfly 0.03 0.05 0.04
20\n hergul 0.26 0.55 0.35 20\n herthr
0.17 0.20 0.19 20\n hoomer 0.00 0.00 0.00
4\n hoowar 0.00 0.00 0.00 20\n horgre
0.50 0.23 0.32 13\n horlar 0.08 0.10 0.09
20\n houfin 0.23 0.35 0.28 20\n houspa
0.21 0.45 0.29 20\n houwre 0.00 0.00 0.00
20\n indbun 0.00 0.00 0.00 20\n juntit1
0.00 0.00 0.00 14\n killde 0.32 0.60 0.42
20\n labwoo 0.00 0.00 0.00 17\n larspa

```
0.00      0.00      0.00        19\n      lazbun       0.06      0.05      0.06
19\n      leabit       0.05      0.06      0.05        18\n      leafly
0.24      0.26      0.25        19\n      leasan       0.00      0.00      0.00
15\n      lecthr       0.00      0.00      0.00         4\n      lesgol
0.07      0.05      0.06        20\n      lesnig       0.67      0.25      0.36
8\n      lesyel       0.20      0.35      0.25        20\n      lewwoo
0.00      0.00      0.00         6\n      linspa       0.00      0.00      0.00
20\n      lobcur       0.29      0.15      0.20        13\n      lobdow
0.00      0.00      0.00        15\n      logshr       0.00      0.00      0.00
20\n      lotduc       0.00      0.00      0.00        10\n\n    accuracy
0.12      2473\n   macro avg       0.09      0.11      0.08      2473\nweighted
avg       0.09      0.12      0.09      2473\n'
```

1.9 k-Nearest Neighbours (k = 3)

To gauge how far a non-parametric method could go, we trained a k-nearest neighbours classifier with k = 3. Distance in the engineered feature space often groups calls from the same species, so KNN performed slightly better, achieving 15% accuracy. The result confirmed that meaningful information exists in the tabular features, yet also highlighted KNN's memory inefficiency and sensitivity to irrelevant dimensions.

[38]:
```
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report,␣
↪confusion_matrix
```

[39]:
```
df = pd.read_csv("dataset.csv")
X = df.drop(columns=["filename", "label"])
y = df["label"]

le = LabelEncoder()
y = le.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, stratify=y, random_state=42)
```

[40]:
```
knn = KNeighborsClassifier(n_neighbors=2)
knn.fit(X_train, y_train)
```

[40]:
```
KNeighborsClassifier(n_neighbors=2)
```

[41]:
```
y_pred = knn.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print(f"Test Accuracy: {acc}")
print(classification_report(y_test, y_pred, target_names=le.classes_))
```

```
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(12, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=le.classes_,␣
 ↪yticklabels=le.classes_)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('KNN Confusion Matrix')
plt.show()

'''
Test Accuracy: 0.14516781237363527
              precision    recall  f1-score   support

      aldfly       0.03      0.10      0.04        20
      ameavo       0.18      0.25      0.21         8
      amebit       0.23      0.33      0.27         9
      amecro       0.28      0.35      0.31        20
      amegfi       0.03      0.05      0.04        20
      amekes       0.18      0.20      0.19        15
      amepip       0.14      0.30      0.19        20
      amered       0.11      0.20      0.14        20
      amerob       0.23      0.30      0.26        20
      amewig       0.19      0.43      0.26         7
      amewoo       0.20      0.29      0.24        17
      amtspa       0.04      0.07      0.05        15
      annhum       0.10      0.25      0.14        20
      astfly       0.24      0.40      0.30        20
      baisan       0.35      0.55      0.43        11
      baleag       0.10      0.14      0.12         7
      balori       0.12      0.20      0.15        20
      banswa       0.22      0.35      0.27        20
      barswa       0.16      0.25      0.19        20
      bawwar       0.13      0.20      0.16        15
     belkin1       0.10      0.14      0.11        14
     belspa2       0.10      0.20      0.13        15
      bewwre       0.11      0.20      0.14        20
      bkbcuc       0.25      0.20      0.22        10
     bkbmag1       0.07      0.08      0.08        12
      bkbwar       0.04      0.05      0.04        20
      bkcchi       0.09      0.15      0.11        20
      bkchum       0.25      0.29      0.27        14
      bkhgro       0.12      0.25      0.16        20
      bkpwar       0.23      0.25      0.24        20
      bktspa       0.18      0.26      0.21        19
      blkpho       0.12      0.20      0.15        20
     blugrb1       0.11      0.25      0.16        20
```

```
    blujay      0.14      0.15      0.15       20
    bnhcow      0.03      0.05      0.03       20
    boboli      0.14      0.30      0.19       20
    bongul      0.56      0.62      0.59        8
    brdowl      0.38      0.30      0.33       20
    brebla      0.00      0.00      0.00       10
    brespa      0.14      0.10      0.12       20
    brncre      0.04      0.05      0.04       20
    brnthr      0.15      0.20      0.17       20
    brthum      0.36      0.60      0.45       20
    brwhaw      0.14      0.13      0.14       15
    btbwar      0.16      0.33      0.22       15
    btnwar      0.12      0.10      0.11       20
    btywar      0.08      0.16      0.11       19
    buffle      0.00      0.00      0.00        3
    buggna      0.31      0.20      0.24       20
    buhvir      0.07      0.20      0.11       20
    bulori      0.15      0.20      0.17       20
    bushti      0.20      0.15      0.17       20
    buwtea      0.36      0.45      0.40       11
    buwwar      0.12      0.16      0.14       19
    cacwre      0.24      0.20      0.22       20
    calgul      0.12      0.25      0.17        8
    calqua      0.14      0.11      0.12       18
    camwar      0.12      0.13      0.12       15
    cangoo      0.29      0.35      0.32       20
    canwar      0.09      0.15      0.11       20
    canwre      0.03      0.05      0.04       20
    carwre      0.08      0.05      0.06       20
    casfin      0.05      0.07      0.06       15
   caster1      0.06      0.05      0.05       20
    casvir      0.19      0.33      0.24       18
    cedwax      0.38      0.28      0.32       18
    chispa      0.00      0.00      0.00       20
    chiswi      0.00      0.00      0.00        6
    chswar      0.11      0.15      0.13       20
    chukar      0.33      0.17      0.22        6
    clanut      0.00      0.00      0.00       16
    cliswa      0.00      0.00      0.00       13
    comgol      0.23      0.18      0.20       17
    comgra      0.09      0.10      0.10       20
    comloo      0.14      0.15      0.15       13
    commer      0.26      0.38      0.31       16
    comnig      0.29      0.29      0.29       17
    comrav      0.17      0.20      0.18       20
    comred      0.22      0.20      0.21       20
    comter      0.19      0.20      0.20       20
```

comyel	0.06	0.05	0.06	20
coohaw	0.08	0.06	0.06	18
coshum	0.00	0.00	0.00	4
cowscj1	0.15	0.14	0.15	14
daejun	0.14	0.10	0.12	20
doccor	0.00	0.00	0.00	7
dowwoo	0.12	0.10	0.11	20
dusfly	0.05	0.05	0.05	20
eargre	0.00	0.00	0.00	7
easblu	0.18	0.16	0.17	19
easkin	0.25	0.12	0.17	16
easmea	0.06	0.05	0.06	20
easpho	0.00	0.00	0.00	18
eastow	0.20	0.10	0.13	20
eawpew	0.08	0.05	0.06	20
eucdov	0.27	0.30	0.29	20
eursta	0.00	0.00	0.00	8
evegro	0.20	0.10	0.13	20
fiespa	0.00	0.00	0.00	20
fiscro	0.25	0.24	0.24	17
foxspa	0.17	0.15	0.16	20
gadwal	0.07	0.05	0.06	20
gcrfin	0.20	0.17	0.18	6
gnttow	0.00	0.00	0.00	20
gnwtea	0.38	0.25	0.30	20
gockin	0.27	0.15	0.19	20
gocspa	0.00	0.00	0.00	14
goleag	0.00	0.00	0.00	10
grbher3	0.00	0.00	0.00	14
grcfly	0.07	0.05	0.06	20
greegr	0.12	0.05	0.07	20
greroa	0.18	0.11	0.14	18
greyel	0.43	0.15	0.22	20
grhowl	0.12	0.05	0.07	20
grnher	0.25	0.08	0.12	13
grtgra	0.00	0.00	0.00	20
grycat	0.20	0.19	0.19	16
gryfly	0.12	0.12	0.12	16
haiwoo	0.00	0.00	0.00	20
hamfly	0.14	0.05	0.07	20
hergul	0.67	0.20	0.31	20
herthr	0.23	0.15	0.18	20
hoomer	0.00	0.00	0.00	4
hoowar	0.25	0.05	0.08	20
horgre	0.20	0.08	0.11	13
horlar	0.00	0.00	0.00	20
houfin	0.33	0.15	0.21	20

```
      houspa       0.14      0.05      0.07        20
      houwre       0.40      0.10      0.16        20
      indbun       0.00      0.00      0.00        20
     juntit1       0.00      0.00      0.00        14
      killde       0.50      0.05      0.09        20
      labwoo       1.00      0.06      0.11        17
      larspa       0.20      0.16      0.18        19
      lazbun       0.00      0.00      0.00        19
      leabit       0.00      0.00      0.00        18
      leafly       0.33      0.11      0.16        19
      leasan       0.00      0.00      0.00        15
      lecthr       1.00      0.50      0.67         4
      lesgol       0.00      0.00      0.00        20
      lesnig       1.00      0.12      0.22         8
      lesyel       0.38      0.15      0.21        20
      lewwoo       0.00      0.00      0.00         6
      linspa       0.00      0.00      0.00        20
      lobcur       0.67      0.15      0.25        13
      lobdow       0.33      0.07      0.11        15
      logshr       0.00      0.00      0.00        20
      lotduc       0.00      0.00      0.00        10

    accuracy                           0.15      2473
   macro avg       0.17      0.15      0.14      2473
weighted avg       0.16      0.15      0.14      2473

---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-6-88d70d32ef17> in <cell line: 0>()
      6 cm = confusion_matrix(y_test, y_pred)
      7 plt.figure(figsize=(12, 8))
----> 8 sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=le.
 ↪classes_, yticklabels=le.classes_)
      9 plt.xlabel('Predicted')
     10 plt.ylabel('Actual')

NameError: name 'sns' is not defined
<Figure size 1200x800 with 0 Axes>
'''
```

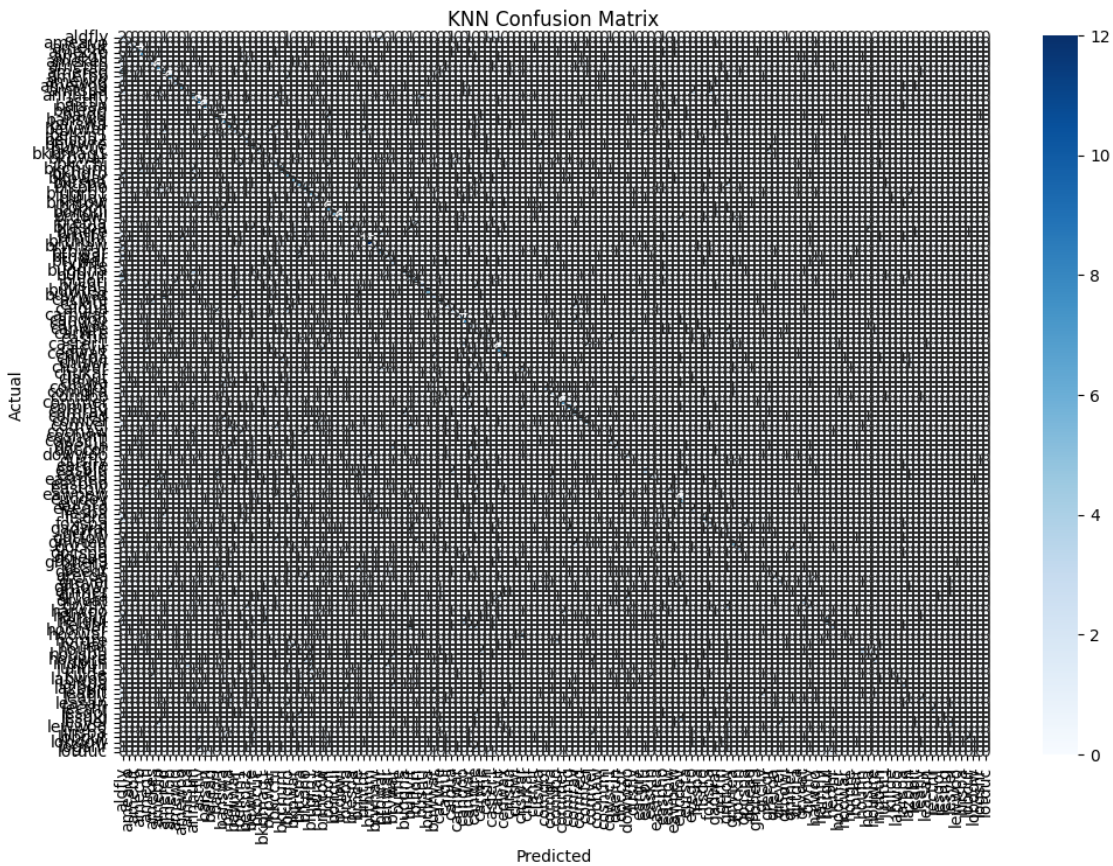
```
Test Accuracy: 0.14516781237363527
              precision    recall  f1-score   support

      aldfly       0.03      0.10      0.04        20
      ameavo       0.18      0.25      0.21         8
      amebit       0.23      0.33      0.27         9
      amecro       0.28      0.35      0.31        20
      amegfi       0.03      0.05      0.04        20
```

amekes	0.18	0.20	0.19	15
amepip	0.14	0.30	0.19	20
amered	0.11	0.20	0.14	20
amerob	0.23	0.30	0.26	20
amewig	0.19	0.43	0.26	7
amewoo	0.20	0.29	0.24	17
amtspa	0.04	0.07	0.05	15
annhum	0.10	0.25	0.14	20
astfly	0.24	0.40	0.30	20
baisan	0.35	0.55	0.43	11
baleag	0.10	0.14	0.12	7
balori	0.12	0.20	0.15	20
banswa	0.22	0.35	0.27	20
barswa	0.16	0.25	0.19	20
bawwar	0.13	0.20	0.16	15
belkin1	0.10	0.14	0.11	14
belspa2	0.10	0.20	0.13	15
bewwre	0.11	0.20	0.14	20
bkbcuc	0.25	0.20	0.22	10
bkbmag1	0.07	0.08	0.08	12
bkbwar	0.04	0.05	0.04	20
bkcchi	0.09	0.15	0.11	20
bkchum	0.25	0.29	0.27	14
bkhgro	0.12	0.25	0.16	20
bkpwar	0.23	0.25	0.24	20
bktspa	0.18	0.26	0.21	19
blkpho	0.12	0.20	0.15	20
blugrb1	0.11	0.25	0.16	20
blujay	0.14	0.15	0.15	20
bnhcow	0.03	0.05	0.03	20
boboli	0.14	0.30	0.19	20
bongul	0.56	0.62	0.59	8
brdowl	0.38	0.30	0.33	20
brebla	0.00	0.00	0.00	10
brespa	0.14	0.10	0.12	20
brncre	0.04	0.05	0.04	20
brnthr	0.15	0.20	0.17	20
brthum	0.36	0.60	0.45	20
brwhaw	0.14	0.13	0.14	15
btbwar	0.16	0.33	0.22	15
btnwar	0.12	0.10	0.11	20
btywar	0.08	0.16	0.11	19
buffle	0.00	0.00	0.00	3
buggna	0.31	0.20	0.24	20
buhvir	0.07	0.20	0.11	20
bulori	0.15	0.20	0.17	20
bushti	0.20	0.15	0.17	20
buwtea	0.36	0.45	0.40	11

buwwar	0.12	0.16	0.14	19
cacwre	0.24	0.20	0.22	20
calgul	0.12	0.25	0.17	8
calqua	0.14	0.11	0.12	18
camwar	0.12	0.13	0.12	15
cangoo	0.29	0.35	0.32	20
canwar	0.09	0.15	0.11	20
canwre	0.03	0.05	0.04	20
carwre	0.08	0.05	0.06	20
casfin	0.05	0.07	0.06	15
caster1	0.06	0.05	0.05	20
casvir	0.19	0.33	0.24	18
cedwax	0.38	0.28	0.32	18
chispa	0.00	0.00	0.00	20
chiswi	0.00	0.00	0.00	6
chswar	0.11	0.15	0.13	20
chukar	0.33	0.17	0.22	6
clanut	0.00	0.00	0.00	16
cliswa	0.00	0.00	0.00	13
comgol	0.23	0.18	0.20	17
comgra	0.09	0.10	0.10	20
comloo	0.14	0.15	0.15	13
commer	0.26	0.38	0.31	16
comnig	0.29	0.29	0.29	17
comrav	0.17	0.20	0.18	20
comred	0.22	0.20	0.21	20
comter	0.19	0.20	0.20	20
comyel	0.06	0.05	0.06	20
coohaw	0.08	0.06	0.06	18
coshum	0.00	0.00	0.00	4
cowscj1	0.15	0.14	0.15	14
daejun	0.14	0.10	0.12	20
doccor	0.00	0.00	0.00	7
dowwoo	0.12	0.10	0.11	20
dusfly	0.05	0.05	0.05	20
eargre	0.00	0.00	0.00	7
easblu	0.18	0.16	0.17	19
easkin	0.25	0.12	0.17	16
easmea	0.06	0.05	0.06	20
easpho	0.00	0.00	0.00	18
eastow	0.20	0.10	0.13	20
eawpew	0.08	0.05	0.06	20
eucdov	0.27	0.30	0.29	20
eursta	0.00	0.00	0.00	8
evegro	0.20	0.10	0.13	20
fiespa	0.00	0.00	0.00	20
fiscro	0.25	0.24	0.24	17
foxspa	0.17	0.15	0.16	20

gadwal	0.07	0.05	0.06	20
gcrfin	0.20	0.17	0.18	6
gnttow	0.00	0.00	0.00	20
gnwtea	0.38	0.25	0.30	20
gockin	0.27	0.15	0.19	20
gocspa	0.00	0.00	0.00	14
goleag	0.00	0.00	0.00	10
grbher3	0.00	0.00	0.00	14
grcfly	0.07	0.05	0.06	20
greegr	0.12	0.05	0.07	20
greroa	0.18	0.11	0.14	18
greyel	0.43	0.15	0.22	20
grhowl	0.12	0.05	0.07	20
grnher	0.25	0.08	0.12	13
grtgra	0.00	0.00	0.00	20
grycat	0.20	0.19	0.19	16
gryfly	0.12	0.12	0.12	16
haiwoo	0.00	0.00	0.00	20
hamfly	0.14	0.05	0.07	20
hergul	0.67	0.20	0.31	20
herthr	0.23	0.15	0.18	20
hoomer	0.00	0.00	0.00	4
hoowar	0.25	0.05	0.08	20
horgre	0.20	0.08	0.11	13
horlar	0.00	0.00	0.00	20
houfin	0.33	0.15	0.21	20
houspa	0.14	0.05	0.07	20
houwre	0.40	0.10	0.16	20
indbun	0.00	0.00	0.00	20
juntit1	0.00	0.00	0.00	14
killde	0.50	0.05	0.09	20
labwoo	1.00	0.06	0.11	17
larspa	0.20	0.16	0.18	19
lazbun	0.00	0.00	0.00	19
leabit	0.00	0.00	0.00	18
leafly	0.33	0.11	0.16	19
leasan	0.00	0.00	0.00	15
lecthr	1.00	0.50	0.67	4
lesgol	0.00	0.00	0.00	20
lesnig	1.00	0.12	0.22	8
lesyel	0.38	0.15	0.21	20
lewwoo	0.00	0.00	0.00	6
linspa	0.00	0.00	0.00	20
lobcur	0.67	0.15	0.25	13
lobdow	0.33	0.07	0.11	15
logshr	0.00	0.00	0.00	20
lotduc	0.00	0.00	0.00	10

```
    accuracy                           0.15      2473
   macro avg       0.17      0.15      0.14      2473
weighted avg       0.16      0.15      0.14      2473
```



[41]:
```
"\nTest Accuracy: 0.14516781237363527\n              precision    recall
f1-score   support\n\n      aldfly       0.03      0.10      0.04         20\n
ameavo       0.18      0.25      0.21          8\n      amebit       0.23
0.33      0.27          9\n      amecro       0.28      0.35      0.31
20\n      amegfi       0.03      0.05      0.04         20\n      amekes
0.18      0.20      0.19         15\n      amepip       0.14      0.30      0.19
20\n      amered       0.11      0.20      0.14         20\n      amerob
0.23      0.30      0.26         20\n      amewig       0.19      0.43      0.26
7\n      amewoo       0.20      0.29      0.24         17\n      amtspa
0.04      0.07      0.05         15\n      annhum       0.10      0.25      0.14
20\n      astfly       0.24      0.40      0.30         20\n      baisan
0.35      0.55      0.43         11\n      baleag       0.10      0.14      0.12
7\n      balori       0.12      0.20      0.15         20\n      banswa
0.22      0.35      0.27         20\n      barswa       0.16      0.25      0.19
20\n      bawwar       0.13      0.20      0.16         15\n      belkin1
```

0.10 0.14 0.11 14\n belspa2 0.10 0.20 0.13
15\n bewwre 0.11 0.20 0.14 20\n bkbcuc
0.25 0.20 0.22 10\n bkbmag1 0.07 0.08 0.08
12\n bkbwar 0.04 0.05 0.04 20\n bkcchi
0.09 0.15 0.11 20\n bkchum 0.25 0.29 0.27
14\n bkhgro 0.12 0.25 0.16 20\n bkpwar
0.23 0.25 0.24 20\n bktspa 0.18 0.26 0.21
19\n blkpho 0.12 0.20 0.15 20\n blugrb1
0.11 0.25 0.16 20\n blujay 0.14 0.15 0.15
20\n bnhcow 0.03 0.05 0.03 20\n boboli
0.14 0.30 0.19 20\n bongul 0.56 0.62 0.59
8\n brdowl 0.38 0.30 0.33 20\n brebla
0.00 0.00 0.00 10\n brespa 0.14 0.10 0.12
20\n brncre 0.04 0.05 0.04 20\n brnthr
0.15 0.20 0.17 20\n brthum 0.36 0.60 0.45
20\n brwhaw 0.14 0.13 0.14 15\n btbwar
0.16 0.33 0.22 15\n btnwar 0.12 0.10 0.11
20\n btywar 0.08 0.16 0.11 19\n buffle
0.00 0.00 0.00 3\n buggna 0.31 0.20 0.24
20\n buhvir 0.07 0.20 0.11 20\n bulori
0.15 0.20 0.17 20\n bushti 0.20 0.15 0.17
20\n buwtea 0.36 0.45 0.40 11\n buwwar
0.12 0.16 0.14 19\n cacwre 0.24 0.20 0.22
20\n calgul 0.12 0.25 0.17 8\n calqua
0.14 0.11 0.12 18\n camwar 0.12 0.13 0.12
15\n cangoo 0.29 0.35 0.32 20\n canwar
0.09 0.15 0.11 20\n canwre 0.03 0.05 0.04
20\n carwre 0.08 0.05 0.06 20\n casfin
0.05 0.07 0.06 15\n caster1 0.06 0.05 0.05
20\n casvir 0.19 0.33 0.24 18\n cedwax
0.38 0.28 0.32 18\n chispa 0.00 0.00 0.00
20\n chiswi 0.00 0.00 0.00 6\n chswar
0.11 0.15 0.13 20\n chukar 0.33 0.17 0.22
6\n clanut 0.00 0.00 0.00 16\n cliswa
0.00 0.00 0.00 13\n comgol 0.23 0.18 0.20
17\n comgra 0.09 0.10 0.10 20\n comloo
0.14 0.15 0.15 13\n commer 0.26 0.38 0.31
16\n comnig 0.29 0.29 0.29 17\n comrav
0.17 0.20 0.18 20\n comred 0.22 0.20 0.21
20\n comter 0.19 0.20 0.20 20\n comyel
0.06 0.05 0.06 20\n coohaw 0.08 0.06 0.06
18\n coshum 0.00 0.00 0.00 4\n cowscj1
0.15 0.14 0.15 14\n daejun 0.14 0.10 0.12
20\n doccor 0.00 0.00 0.00 7\n dowwoo
0.12 0.10 0.11 20\n dusfly 0.05 0.05 0.05
20\n eargre 0.00 0.00 0.00 7\n easblu
0.18 0.16 0.17 19\n easkin 0.25 0.12 0.17

16\n easmea 0.06 0.05 0.06 20\n easpho
0.00 0.00 0.00 18\n eastow 0.20 0.10 0.13
20\n eawpew 0.08 0.05 0.06 20\n eucdov
0.27 0.30 0.29 20\n eursta 0.00 0.00 0.00
8\n evegro 0.20 0.10 0.13 20\n fiespa
0.00 0.00 0.00 20\n fiscro 0.25 0.24 0.24
17\n foxspa 0.17 0.15 0.16 20\n gadwal
0.07 0.05 0.06 20\n gcrfin 0.20 0.17 0.18
6\n gnttow 0.00 0.00 0.00 20\n gnwtea
0.38 0.25 0.30 20\n gockin 0.27 0.15 0.19
20\n gocspa 0.00 0.00 0.00 14\n goleag
0.00 0.00 0.00 10\n grbher3 0.00 0.00 0.00
14\n grcfly 0.07 0.05 0.06 20\n greegr
0.12 0.05 0.07 20\n greroa 0.18 0.11 0.14
18\n greyel 0.43 0.15 0.22 20\n grhowl
0.12 0.05 0.07 20\n grnher 0.25 0.08 0.12
13\n grtgra 0.00 0.00 0.00 20\n grycat
0.20 0.19 0.19 16\n gryfly 0.12 0.12 0.12
16\n haiwoo 0.00 0.00 0.00 20\n hamfly
0.14 0.05 0.07 20\n hergul 0.67 0.20 0.31
20\n herthr 0.23 0.15 0.18 20\n hoomer
0.00 0.00 0.00 4\n hoowar 0.25 0.05 0.08
20\n horgre 0.20 0.08 0.11 13\n horlar
0.00 0.00 0.00 20\n houfin 0.33 0.15 0.21
20\n houspa 0.14 0.05 0.07 20\n houwre
0.40 0.10 0.16 20\n indbun 0.00 0.00 0.00
20\n juntit1 0.00 0.00 0.00 14\n killde
0.50 0.05 0.09 20\n labwoo 1.00 0.06 0.11
17\n larspa 0.20 0.16 0.18 19\n lazbun
0.00 0.00 0.00 19\n leabit 0.00 0.00 0.00
18\n leafly 0.33 0.11 0.16 19\n leasan
0.00 0.00 0.00 15\n lecthr 1.00 0.50 0.67
4\n lesgol 0.00 0.00 0.00 20\n lesnig
1.00 0.12 0.22 8\n lesyel 0.38 0.15 0.21
20\n lewwoo 0.00 0.00 0.00 6\n linspa
0.00 0.00 0.00 20\n lobcur 0.67 0.15 0.25
13\n lobdow 0.33 0.07 0.11 15\n logshr
0.00 0.00 0.00 20\n lotduc 0.00 0.00 0.00
10\n\n accuracy 0.15 2473\n macro avg
0.17 0.15 0.14 2473\nweighted avg 0.16 0.15 0.14
2473\n\n---
---\nNameError Traceback (most recent call
last)\n<ipython-input-6-88d70d32ef17> in <cell line: 0>()\n 6 cm =
confusion_matrix(y_test, y_pred)\n 7 plt.figure(figsize=(12, 8))\n----> 8
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=le.classes_,
yticklabels=le.classes_)\n 9 plt.xlabel('Predicted')\n 10
plt.ylabel('Actual')\n\nNameError: name 'sns' is not defined\n<Figure size

```
1200x800 with 0 Axes>\n"
```

1.10 PyTorch Model (Best Performing)

After validating our feature engineering and training setup in TensorFlow, we transitioned to PyTorch for more granular control over the training process. We implemented a simple fully connected network with:

- One hidden layer using 80 neurons and ReLU activation
- Dropout (0.3) after the hidden layer to prevent overfitting
- A final output layer with softmax activation
- CrossEntropyLoss as the loss function
- Adam optimizer with weight decay

To find the best configuration, we wrapped the PyTorch training loop in an Optuna study. The tuning parameters included:

- Learning rate (0.00001 to 0.01)
- Batch size (8, 16, 32, 64)
- Hidden layer size (32–128)
- Number of hidden layers (1–3)
- Optimizer: Adam or SGD

We ran 50 trials with 100 epochs each, using early stopping based on validation loss. Unlike our earlier models, this setup allowed us to dynamically construct different model architectures depending on each trial's configuration.

- The best-performing PyTorch model used:
- 1 hidden layer with 80 neurons
- ReLU activation
- Learning rate of 0.00158
- Batch size of 32
- Adam optimizer

This configuration reached a test accuracy of 17%, the highest in our entire project. While still far from ideal due to class imbalance and the complexity of the task, this performance confirms the value of PyTorch for flexible model design and high-performance training loops. It also demonstrates that our feature set carries meaningful patterns that deep learning models can leverage effectively.

[42]:
```
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import TensorDataset, DataLoader
import optuna
```

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix,␣
↪classification_report
import matplotlib.pyplot as plt
import seaborn as sns
```

[43]:
```python
df = pd.read_csv("dataset.csv")
X = df.drop(columns=["filename", "label"])
y = df["label"]

le = LabelEncoder()
y = le.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X).astype(np.float32)
y = y.astype(np.int64)

X_trainval, X_test, y_trainval, y_test = train_test_split(
    X_scaled, y, test_size=0.2, stratify=y, random_state=42)
X_train, X_val, y_train, y_val = train_test_split(
    X_trainval, y_trainval, test_size=0.2, stratify=y_trainval, random_state=42)
```

[44]:
```python
class BirdClassifier(nn.Module):
    def __init__(self, input_dim, hidden_dim, n_layers, output_dim):
        super().__init__()
        layers = [nn.Linear(input_dim, hidden_dim), nn.ReLU()]
        for _ in range(n_layers - 1):
            layers += [nn.Linear(hidden_dim, hidden_dim), nn.ReLU()]
        layers.append(nn.Dropout(0.5))
        layers.append(nn.Linear(hidden_dim, output_dim))
        self.network = nn.Sequential(*layers)

    def forward(self, x):
        return self.network(x)
```

[45]:
```python
# optuna
def objective(trial):
    hidden_dim = trial.suggest_int("hidden_dim", 32, 128)
    n_layers = trial.suggest_int("n_layers", 1, 20)
    lr = trial.suggest_loguniform("lr", 1e-5, 1e-1)
    batch_size = trial.suggest_categorical("batch_size", [16, 32, 64, 128])

    train_dataset = TensorDataset(torch.from_numpy(X_train), torch.
↪from_numpy(y_train))
```

```
    val_dataset = TensorDataset(torch.from_numpy(X_val), torch.
↪from_numpy(y_val))
    train_loader = DataLoader(train_dataset, batch_size=batch_size,␣
↪shuffle=True)
    val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)

    model = BirdClassifier(X_train.shape[1], hidden_dim, n_layers, len(np.
↪unique(y)))
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model.to(device)
    optimizer = optim.Adam(model.parameters(), lr=lr)
    criterion = nn.CrossEntropyLoss()

    for epoch in range(20):
        model.train()
        for xb, yb in train_loader:
            xb, yb = xb.to(device), yb.to(device)
            optimizer.zero_grad()
            loss = criterion(model(xb), yb)
            loss.backward()
            optimizer.step()

    model.eval()
    preds, labels = [], []
    with torch.no_grad():
        for xb, yb in val_loader:
            xb = xb.to(device)
            output = model(xb)
            preds.extend(torch.argmax(output, axis=1).cpu().numpy())
            labels.extend(yb.numpy())
    return accuracy_score(labels, preds)
```

[46]:
```
# Run Optuna study
study = optuna.create_study(direction="maximize")
study.optimize(objective, n_trials=50)
best_params = study.best_trial.params
print("Best hyperparameters:", best_params)

#Best hyperparameters: {'hidden_dim': 80, 'n_layers': 1, 'lr': 0.
↪0015846805259486921, 'batch_size': 32}
```

```
[I 2025-05-22 21:54:19,994] A new study created in memory with name: no-
name-3fc4def9-1fed-48f5-8946-985911637261
[I 2025-05-22 21:54:47,893] Trial 0 finished with value: 0.10308236483072258 and
parameters: {'hidden_dim': 52, 'n_layers': 3, 'lr': 0.002568017166774806,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:54:57,459] Trial 1 finished with value: 0.009095502779181405
```

```
and parameters: {'hidden_dim': 62, 'n_layers': 11, 'lr': 7.994219247640015e-05,
'batch_size': 128}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:55:27,503] Trial 2 finished with value: 0.027791814047498736
and parameters: {'hidden_dim': 112, 'n_layers': 16, 'lr': 0.0006167277509475986,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:55:33,187] Trial 3 finished with value: 0.00808489135927236 and
parameters: {'hidden_dim': 48, 'n_layers': 4, 'lr': 0.06951882785687288,
'batch_size': 128}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:55:44,291] Trial 4 finished with value: 0.02425467407781708 and
parameters: {'hidden_dim': 53, 'n_layers': 15, 'lr': 0.005385000785426969,
'batch_size': 128}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:55:49,520] Trial 5 finished with value: 0.0404244567963618 and
parameters: {'hidden_dim': 104, 'n_layers': 1, 'lr': 6.004201131610597e-05,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:56:18,444] Trial 6 finished with value: 0.00808489135927236 and
parameters: {'hidden_dim': 65, 'n_layers': 4, 'lr': 0.012313585049475082,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:56:48,655] Trial 7 finished with value: 0.02425467407781708 and
parameters: {'hidden_dim': 44, 'n_layers': 13, 'lr': 0.0025236769710173857,
'batch_size': 32}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:57:04,964] Trial 8 finished with value: 0.01616978271854472 and
parameters: {'hidden_dim': 42, 'n_layers': 15, 'lr': 9.512293689687046e-05,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:57:15,652] Trial 9 finished with value: 0.00808489135927236 and
parameters: {'hidden_dim': 108, 'n_layers': 1, 'lr': 0.05999713299771257,
'batch_size': 32}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:58:04,592] Trial 10 finished with value: 0.03840323395654371
and parameters: {'hidden_dim': 83, 'n_layers': 8, 'lr': 0.0005210233726852421,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 21:58:09,825] Trial 11 finished with value: 0.01010611419909045
and parameters: {'hidden_dim': 87, 'n_layers': 1, 'lr': 1.0827714007055108e-05,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:00:02,090] Trial 12 finished with value: 0.017180394138453764
and parameters: {'hidden_dim': 99, 'n_layers': 20, 'lr': 0.00010062409707895873,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:00:46,495] Trial 13 finished with value: 0.00808489135927236
and parameters: {'hidden_dim': 128, 'n_layers': 7, 'lr': 1.276827361069749e-05,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:00:55,338] Trial 14 finished with value: 0.07579585649317837
and parameters: {'hidden_dim': 66, 'n_layers': 4, 'lr': 0.0003220081011678429,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:01:03,886] Trial 15 finished with value: 0.07225871652349672
and parameters: {'hidden_dim': 32, 'n_layers': 5, 'lr': 0.002412304134070082,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:01:46,924] Trial 16 finished with value: 0.027791814047498736
and parameters: {'hidden_dim': 70, 'n_layers': 8, 'lr': 0.00028841331414976887,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:02:08,620] Trial 17 finished with value: 0.00808489135927236
```

```
and parameters: {'hidden_dim': 74, 'n_layers': 6, 'lr': 0.015421054539051503,
'batch_size': 32}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:02:22,758] Trial 18 finished with value: 0.027286508337544216
and parameters: {'hidden_dim': 54, 'n_layers': 10, 'lr': 0.0017935450554440594,
'batch_size': 64}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:02:42,363] Trial 19 finished with value: 0.0828701364325417 and
parameters: {'hidden_dim': 33, 'n_layers': 3, 'lr': 0.0003370496134258279,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:03:03,677] Trial 20 finished with value: 0.06417382516422436
and parameters: {'hidden_dim': 33, 'n_layers': 3, 'lr': 0.005986857093323584,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:03:25,036] Trial 21 finished with value: 0.09701869631126832
and parameters: {'hidden_dim': 59, 'n_layers': 3, 'lr': 0.0002453158697896104,
'batch_size': 16}. Best is trial 0 with value: 0.10308236483072258.
[I 2025-05-22 22:03:45,600] Trial 22 finished with value: 0.10358767054067711
and parameters: {'hidden_dim': 55, 'n_layers': 3, 'lr': 0.0002319591774362952,
'batch_size': 16}. Best is trial 22 with value: 0.10358767054067711.
[I 2025-05-22 22:04:17,314] Trial 23 finished with value: 0.01414855987872663
and parameters: {'hidden_dim': 55, 'n_layers': 6, 'lr': 3.5205140344508536e-05,
'batch_size': 16}. Best is trial 22 with value: 0.10358767054067711.
[I 2025-05-22 22:04:35,294] Trial 24 finished with value: 0.13542193026781202
and parameters: {'hidden_dim': 60, 'n_layers': 2, 'lr': 0.0010682607865728747,
'batch_size': 16}. Best is trial 24 with value: 0.13542193026781202.
[I 2025-05-22 22:04:55,067] Trial 25 finished with value: 0.13643254168772107
and parameters: {'hidden_dim': 77, 'n_layers': 2, 'lr': 0.0018090876779505395,
'batch_size': 16}. Best is trial 25 with value: 0.13643254168772107.
[I 2025-05-22 22:05:10,145] Trial 26 finished with value: 0.14300151591712987
and parameters: {'hidden_dim': 77, 'n_layers': 1, 'lr': 0.000840887582831203,
'batch_size': 16}. Best is trial 26 with value: 0.14300151591712987.
[I 2025-05-22 22:05:25,490] Trial 27 finished with value: 0.14401212733703891
and parameters: {'hidden_dim': 92, 'n_layers': 1, 'lr': 0.00100203145140095,
'batch_size': 16}. Best is trial 27 with value: 0.14401212733703891.
[I 2025-05-22 22:05:40,536] Trial 28 finished with value: 0.1435068216270844 and
parameters: {'hidden_dim': 92, 'n_layers': 1, 'lr': 0.0011862884314888709,
'batch_size': 16}. Best is trial 27 with value: 0.14401212733703891.
[I 2025-05-22 22:05:55,980] Trial 29 finished with value: 0.1505811015664477 and
parameters: {'hidden_dim': 94, 'n_layers': 1, 'lr': 0.0012014770073852973,
'batch_size': 16}. Best is trial 29 with value: 0.1505811015664477.
[I 2025-05-22 22:07:04,234] Trial 30 finished with value: 0.00808489135927236
and parameters: {'hidden_dim': 93, 'n_layers': 10, 'lr': 0.004805075526362826,
'batch_size': 16}. Best is trial 29 with value: 0.1505811015664477.
[I 2025-05-22 22:07:19,349] Trial 31 finished with value: 0.14300151591712987
and parameters: {'hidden_dim': 90, 'n_layers': 1, 'lr': 0.0009722821339684958,
'batch_size': 16}. Best is trial 29 with value: 0.1505811015664477.
[I 2025-05-22 22:07:39,629] Trial 32 finished with value: 0.16119252147549268
and parameters: {'hidden_dim': 99, 'n_layers': 2, 'lr': 0.0011177697036302936,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:07:47,270] Trial 33 finished with value: 0.1147043961596766 and
```

```
parameters: {'hidden_dim': 115, 'n_layers': 5, 'lr': 0.0014278575445440473,
'batch_size': 128}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:08:07,314] Trial 34 finished with value: 0.14401212733703891
and parameters: {'hidden_dim': 98, 'n_layers': 2, 'lr': 0.0005571443092770476,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:08:13,739] Trial 35 finished with value: 0.09146033350176858
and parameters: {'hidden_dim': 97, 'n_layers': 5, 'lr': 0.0004892377952262613,
'batch_size': 128}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:08:33,525] Trial 36 finished with value: 0.11419909044972208
and parameters: {'hidden_dim': 115, 'n_layers': 2, 'lr': 0.00014909124197875999,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:08:51,833] Trial 37 finished with value: 0.07781707933299646
and parameters: {'hidden_dim': 102, 'n_layers': 4, 'lr': 0.004561329870697131,
'batch_size': 32}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:09:05,733] Trial 38 finished with value: 0.025265285497726123
and parameters: {'hidden_dim': 108, 'n_layers': 12, 'lr': 0.0006308780962046102,
'batch_size': 128}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:11:34,258] Trial 39 finished with value: 0.00808489135927236
and parameters: {'hidden_dim': 124, 'n_layers': 19, 'lr': 0.009736233458172651,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:11:55,572] Trial 40 finished with value: 0.1278423446184942 and
parameters: {'hidden_dim': 83, 'n_layers': 2, 'lr': 0.003368182409322662,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:12:15,018] Trial 41 finished with value: 0.15613946437594745
and parameters: {'hidden_dim': 94, 'n_layers': 2, 'lr': 0.0011129442107155106,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:12:38,453] Trial 42 finished with value: 0.15007579585649317
and parameters: {'hidden_dim': 97, 'n_layers': 3, 'lr': 0.000778901676006415,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:13:09,635] Trial 43 finished with value: 0.0965133906013138 and
parameters: {'hidden_dim': 87, 'n_layers': 4, 'lr': 0.002648418156901688,
'batch_size': 16}. Best is trial 32 with value: 0.16119252147549268.
[I 2025-05-22 22:13:29,635] Trial 44 finished with value: 0.16523496715512886
and parameters: {'hidden_dim': 103, 'n_layers': 2, 'lr': 0.0007308630274354008,
'batch_size': 16}. Best is trial 44 with value: 0.16523496715512886.
[I 2025-05-22 22:13:42,284] Trial 45 finished with value: 0.10005053057099546
and parameters: {'hidden_dim': 104, 'n_layers': 3, 'lr': 0.00015300354155917272,
'batch_size': 32}. Best is trial 44 with value: 0.16523496715512886.
[I 2025-05-22 22:14:32,898] Trial 46 finished with value: 0.05962607377463365
and parameters: {'hidden_dim': 109, 'n_layers': 7, 'lr': 0.0007460579449983475,
'batch_size': 16}. Best is trial 44 with value: 0.16523496715512886.
[I 2025-05-22 22:15:09,480] Trial 47 finished with value: 0.10813542193026782
and parameters: {'hidden_dim': 120, 'n_layers': 5, 'lr': 0.0004624526017353928,
'batch_size': 16}. Best is trial 44 with value: 0.16523496715512886.
[I 2025-05-22 22:15:13,985] Trial 48 finished with value: 0.14502273875694796
and parameters: {'hidden_dim': 103, 'n_layers': 2, 'lr': 0.00155988622271322,
'batch_size': 128}. Best is trial 44 with value: 0.16523496715512886.
[I 2025-05-22 22:16:56,815] Trial 49 finished with value: 0.00808489135927236
```

```
and parameters: {'hidden_dim': 95, 'n_layers': 17, 'lr': 0.0030513777118366145,
'batch_size': 16}. Best is trial 44 with value: 0.16523496715512886.

Best hyperparameters: {'hidden_dim': 103, 'n_layers': 2, 'lr':
0.0007308630274354008, 'batch_size': 16}
```

[47]:
```
# Retrain model with best hyperparameters and evaluate
final_model = BirdClassifier(
    input_dim=X_train.shape[1],
    hidden_dim=best_params['hidden_dim'],
    n_layers=best_params['n_layers'],
    output_dim=len(np.unique(y))
)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
final_model.to(device)

optimizer = optim.Adam(final_model.parameters(), lr=best_params['lr'])
criterion = nn.CrossEntropyLoss()

train_dataset = TensorDataset(torch.from_numpy(X_train), torch.
 ↪from_numpy(y_train))
train_loader = DataLoader(train_dataset, batch_size=best_params['batch_size'],␣
 ↪shuffle=True)
test_dataset = TensorDataset(torch.from_numpy(X_test), torch.from_numpy(y_test))
test_loader = DataLoader(test_dataset, batch_size=best_params['batch_size'],␣
 ↪shuffle=False)

for epoch in range(100):
    final_model.train()
    for xb, yb in train_loader:
        xb, yb = xb.to(device), yb.to(device)
        optimizer.zero_grad()
        loss = criterion(final_model(xb), yb)
        loss.backward()
        optimizer.step()
```

[48]:
```
final_model.eval()
y_true, y_pred = [], []
with torch.no_grad():
    for xb, yb in test_loader:
        xb = xb.to(device)
        outputs = final_model(xb)
        y_true.extend(yb.numpy())
        y_pred.extend(torch.argmax(outputs, dim=1).cpu().numpy())

acc = accuracy_score(y_true, y_pred)
print(f"Test Accuracy: {acc}")
print(classification_report(y_true, y_pred, target_names=le.classes_))
```

```
cm = confusion_matrix(y_true, y_pred)
plt.figure(figsize=(12, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=le.classes_,␣
 ↪yticklabels=le.classes_)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

'''
Test Accuracy: 0.1678123736352608
              precision    recall  f1-score   support

      aldfly       0.08      0.05      0.06        20
      ameavo       0.17      0.12      0.14         8
      amebit       0.00      0.00      0.00         9
      amecro       0.20      0.45      0.27        20
      amegfi       0.20      0.05      0.08        20
      amekes       0.40      0.13      0.20        15
      amepip       0.08      0.10      0.09        20
      amered       0.20      0.10      0.13        20
      amerob       0.19      0.35      0.25        20
      amewig       0.43      0.43      0.43         7
      amewoo       0.50      0.24      0.32        17
      amtspa       0.00      0.00      0.00        15
      annhum       0.08      0.15      0.11        20
      astfly       0.16      0.30      0.21        20
      baisan       0.00      0.00      0.00        11
      baleag       0.00      0.00      0.00         7
      balori       0.19      0.20      0.20        20
      banswa       0.24      0.20      0.22        20
      barswa       0.24      0.40      0.30        20
      bawwar       0.07      0.07      0.07        15
     belkin1       0.00      0.00      0.00        14
     belspa2       0.30      0.20      0.24        15
      bewwre       0.00      0.00      0.00        20
      bkbcuc       1.00      0.10      0.18        10
     bkbmag1       0.00      0.00      0.00        12
      bkbwar       0.07      0.15      0.10        20
      bkcchi       0.06      0.10      0.07        20
      bkchum       0.27      0.64      0.38        14
      bkhgro       0.00      0.00      0.00        20
      bkpwar       0.29      0.65      0.40        20
      bktspa       0.25      0.11      0.15        19
      blkpho       0.25      0.15      0.19        20
     blugrb1       0.00      0.00      0.00        20
```

```
    blujay      0.05      0.05      0.05       20
    bnhcow      0.00      0.00      0.00       20
    boboli      0.19      0.20      0.20       20
    bongul      0.20      0.25      0.22        8
    brdowl      0.13      0.10      0.11       20
    brebla      0.00      0.00      0.00       10
    brespa      0.09      0.10      0.09       20
    brncre      0.00      0.00      0.00       20
    brnthr      0.13      0.20      0.16       20
    brthum      0.54      0.65      0.59       20
    brwhaw      0.33      0.07      0.11       15
    btbwar      0.00      0.00      0.00       15
    btnwar      0.00      0.00      0.00       20
    btywar      0.09      0.05      0.07       19
    buffle      0.00      0.00      0.00        3
    buggna      0.33      0.15      0.21       20
    buhvir      0.07      0.10      0.08       20
    bulori      0.20      0.35      0.25       20
    bushti      0.15      0.35      0.21       20
    buwtea      0.29      0.18      0.22       11
    buwwar      0.15      0.11      0.12       19
    cacwre      0.18      0.35      0.24       20
    calgul      0.17      0.12      0.14        8
    calqua      0.31      0.28      0.29       18
    camwar      0.14      0.13      0.14       15
    cangoo      0.31      0.65      0.42       20
    canwar      0.00      0.00      0.00       20
    canwre      0.08      0.10      0.09       20
    carwre      0.00      0.00      0.00       20
    casfin      0.00      0.00      0.00       15
   caster1      0.15      0.20      0.17       20
    casvir      0.22      0.22      0.22       18
    cedwax      0.14      0.17      0.15       18
    chispa      0.09      0.10      0.09       20
    chiswi      0.00      0.00      0.00        6
    chswar      0.06      0.20      0.09       20
    chukar      0.00      0.00      0.00        6
    clanut      0.24      0.44      0.31       16
    cliswa      0.07      0.08      0.07       13
    comgol      0.36      0.29      0.32       17
    comgra      0.06      0.05      0.06       20
    comloo      0.29      0.15      0.20       13
    commer      0.19      0.31      0.24       16
    comnig      0.00      0.00      0.00       17
    comrav      0.18      0.40      0.25       20
    comred      0.24      0.25      0.24       20
    comter      0.34      0.50      0.41       20
```

comyel	0.00	0.00	0.00	20
coohaw	0.00	0.00	0.00	18
coshum	0.00	0.00	0.00	4
cowscj1	0.00	0.00	0.00	14
daejun	0.21	0.30	0.25	20
doccor	0.00	0.00	0.00	7
dowwoo	0.29	0.10	0.15	20
dusfly	0.09	0.10	0.09	20
eargre	0.00	0.00	0.00	7
easblu	0.00	0.00	0.00	19
easkin	0.27	0.19	0.22	16
easmea	0.00	0.00	0.00	20
easpho	0.00	0.00	0.00	18
eastow	0.17	0.10	0.12	20
eawpew	0.50	0.10	0.17	20
eucdov	0.31	0.50	0.38	20
eursta	0.00	0.00	0.00	8
evegro	0.02	0.05	0.03	20
fiespa	0.20	0.05	0.08	20
fiscro	0.10	0.12	0.11	17
foxspa	0.14	0.20	0.17	20
gadwal	0.10	0.10	0.10	20
gcrfin	0.00	0.00	0.00	6
gnttow	0.09	0.30	0.13	20
gnwtea	0.31	0.55	0.40	20
gockin	0.21	0.40	0.28	20
gocspa	0.00	0.00	0.00	14
goleag	0.00	0.00	0.00	10
grbher3	0.00	0.00	0.00	14
grcfly	0.14	0.15	0.15	20
greegr	0.21	0.15	0.18	20
greroa	0.10	0.17	0.13	18
greyel	0.40	0.20	0.27	20
grhowl	0.24	0.20	0.22	20
grnher	0.11	0.08	0.09	13
grtgra	0.12	0.05	0.07	20
grycat	0.25	0.06	0.10	16
gryfly	0.07	0.19	0.10	16
haiwoo	0.00	0.00	0.00	20
hamfly	0.12	0.15	0.13	20
hergul	0.38	0.60	0.46	20
herthr	0.16	0.40	0.23	20
hoomer	0.00	0.00	0.00	4
hoowar	0.11	0.05	0.07	20
horgre	0.50	0.15	0.24	13
horlar	0.10	0.20	0.13	20
houfin	0.23	0.45	0.30	20

```
      houspa       0.15      0.40      0.22        20
      houwre       0.08      0.05      0.06        20
      indbun       0.33      0.05      0.09        20
     juntit1       0.33      0.14      0.20        14
      killde       0.54      0.65      0.59        20
      labwoo       0.00      0.00      0.00        17
      larspa       0.00      0.00      0.00        19
      lazbun       0.19      0.16      0.17        19
      leabit       0.04      0.06      0.05        18
      leafly       0.22      0.11      0.14        19
      leasan       0.20      0.07      0.10        15
      lecthr       0.00      0.00      0.00         4
      lesgol       0.10      0.05      0.07        20
      lesnig       0.44      0.50      0.47         8
      lesyel       0.29      0.40      0.33        20
      lewwoo       0.00      0.00      0.00         6
      linspa       0.00      0.00      0.00        20
      lobcur       0.12      0.15      0.14        13
      lobdow       0.00      0.00      0.00        15
      logshr       0.11      0.10      0.11        20
      lotduc       0.50      0.40      0.44        10

    accuracy                           0.17      2473
   macro avg       0.15      0.16      0.14      2473
weighted avg       0.16      0.17      0.15      2473

/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565:
 ↪ UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in␣
 ↪labels with no predicted samples. Use `zero_division` parameter to control␣
 ↪this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565:
 ↪ UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in␣
 ↪labels with no predicted samples. Use `zero_division` parameter to control␣
 ↪this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.11/dist-packages/sklearn/metrics/_classification.py:1565:
 ↪ UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in␣
 ↪labels with no predicted samples. Use `zero_division` parameter to control␣
 ↪this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))

'''
```

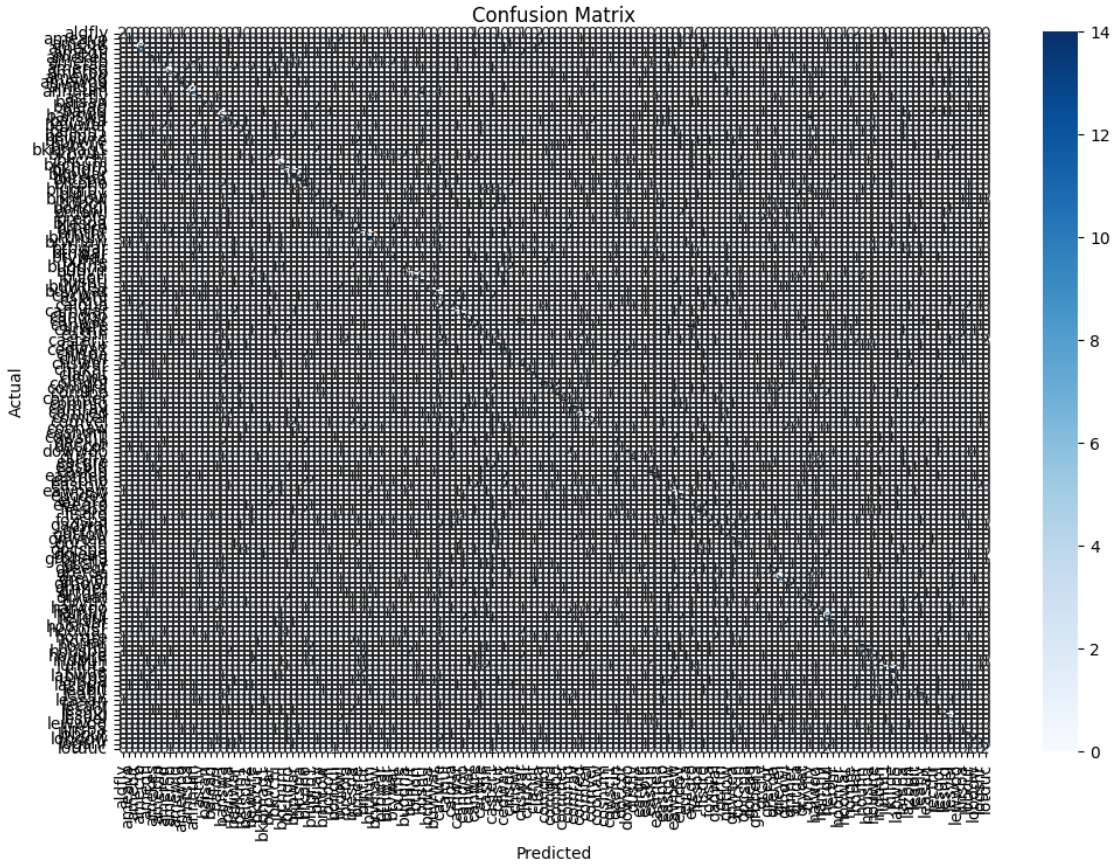
```
Test Accuracy: 0.1819652244237768
              precision    recall  f1-score   support
```

aldfly	0.14	0.10	0.12	20
ameavo	0.14	0.12	0.13	8
amebit	0.00	0.00	0.00	9
amecro	0.30	0.45	0.36	20
amegfi	0.00	0.00	0.00	20
amekes	0.21	0.33	0.26	15
amepip	0.00	0.00	0.00	20
amered	0.16	0.15	0.15	20
amerob	0.27	0.40	0.32	20
amewig	0.50	0.43	0.46	7
amewoo	0.29	0.24	0.26	17
amtspa	0.20	0.13	0.16	15
annhum	0.17	0.40	0.24	20
astfly	0.17	0.25	0.20	20
baisan	0.25	0.18	0.21	11
baleag	0.33	0.14	0.20	7
balori	0.12	0.10	0.11	20
banswa	0.24	0.45	0.31	20
barswa	0.20	0.25	0.22	20
bawwar	0.15	0.13	0.14	15
belkin1	0.20	0.14	0.17	14
belspa2	0.27	0.20	0.23	15
bewwre	0.05	0.05	0.05	20
bkbcuc	0.00	0.00	0.00	10
bkbmag1	0.00	0.00	0.00	12
bkbwar	0.13	0.10	0.11	20
bkcchi	0.00	0.00	0.00	20
bkchum	0.28	0.57	0.37	14
bkhgro	0.21	0.25	0.23	20
bkpwar	0.37	0.65	0.47	20
bktspa	0.29	0.21	0.24	19
blkpho	0.17	0.20	0.18	20
blugrb1	0.06	0.05	0.06	20
blujay	0.12	0.10	0.11	20
bnhcow	0.00	0.00	0.00	20
boboli	0.21	0.15	0.18	20
bongul	0.25	0.25	0.25	8
brdowl	0.22	0.30	0.26	20
brebla	0.00	0.00	0.00	10
brespa	0.04	0.05	0.05	20
brncre	0.07	0.10	0.08	20
brnthr	0.21	0.25	0.23	20
brthum	0.44	0.60	0.51	20
brwhaw	0.17	0.07	0.10	15
btbwar	0.08	0.07	0.07	15
btnwar	0.11	0.05	0.07	20
btywar	0.12	0.16	0.14	19
buffle	0.00	0.00	0.00	3

buggna	0.11	0.10	0.11	20
buhvir	0.10	0.05	0.07	20
bulori	0.32	0.50	0.39	20
bushti	0.19	0.25	0.22	20
buwtea	0.18	0.18	0.18	11
buwwar	0.12	0.11	0.11	19
cacwre	0.21	0.35	0.26	20
calgul	0.00	0.00	0.00	8
calqua	0.13	0.11	0.12	18
camwar	0.14	0.20	0.16	15
cangoo	0.45	0.70	0.55	20
canwar	0.14	0.10	0.12	20
canwre	0.10	0.10	0.10	20
carwre	0.06	0.05	0.06	20
casfin	0.06	0.07	0.06	15
caster1	0.03	0.05	0.04	20
casvir	0.29	0.28	0.29	18
cedwax	0.26	0.33	0.29	18
chispa	0.06	0.05	0.05	20
chiswi	0.00	0.00	0.00	6
chswar	0.12	0.20	0.15	20
chukar	0.67	0.33	0.44	6
clanut	0.19	0.31	0.24	16
cliswa	0.00	0.00	0.00	13
comgol	0.40	0.35	0.38	17
comgra	0.17	0.15	0.16	20
comloo	0.33	0.31	0.32	13
commer	0.31	0.25	0.28	16
comnig	0.00	0.00	0.00	17
comrav	0.24	0.30	0.27	20
comred	0.11	0.15	0.13	20
comter	0.42	0.50	0.45	20
comyel	0.09	0.10	0.10	20
coohaw	0.00	0.00	0.00	18
coshum	0.00	0.00	0.00	4
cowscj1	0.00	0.00	0.00	14
daejun	0.12	0.15	0.13	20
doccor	0.20	0.14	0.17	7
dowwoo	0.17	0.15	0.16	20
dusfly	0.21	0.20	0.21	20
eargre	0.00	0.00	0.00	7
easblu	0.00	0.00	0.00	19
easkin	0.33	0.25	0.29	16
easmea	0.29	0.20	0.24	20
easpho	0.00	0.00	0.00	18
eastow	0.08	0.05	0.06	20
eawpew	0.08	0.10	0.09	20
eucdov	0.47	0.70	0.56	20

eursta	0.00	0.00	0.00	8
evegro	0.10	0.15	0.12	20
fiespa	0.18	0.20	0.19	20
fiscro	0.11	0.12	0.11	17
foxspa	0.12	0.10	0.11	20
gadwal	0.11	0.10	0.11	20
gcrfin	0.17	0.17	0.17	6
gnttow	0.11	0.10	0.10	20
gnwtea	0.18	0.25	0.21	20
gockin	0.23	0.30	0.26	20
gocspa	0.00	0.00	0.00	14
goleag	1.00	0.10	0.18	10
grbher3	0.00	0.00	0.00	14
grcfly	0.00	0.00	0.00	20
greegr	0.20	0.15	0.17	20
greroa	0.11	0.17	0.13	18
greyel	0.32	0.40	0.36	20
grhowl	0.20	0.15	0.17	20
grnher	0.00	0.00	0.00	13
grtgra	0.09	0.05	0.06	20
grycat	0.12	0.12	0.12	16
gryfly	0.12	0.31	0.17	16
haiwoo	0.08	0.10	0.09	20
hamfly	0.17	0.20	0.19	20
hergul	0.36	0.45	0.40	20
herthr	0.25	0.30	0.27	20
hoomer	0.00	0.00	0.00	4
hoowar	0.09	0.05	0.06	20
horgre	0.38	0.23	0.29	13
horlar	0.12	0.15	0.13	20
houfin	0.28	0.25	0.26	20
houspa	0.29	0.25	0.27	20
houwre	0.16	0.15	0.15	20
indbun	0.17	0.10	0.12	20
juntit1	0.33	0.07	0.12	14
killde	0.48	0.65	0.55	20
labwoo	0.14	0.06	0.08	17
larspa	0.00	0.00	0.00	19
lazbun	0.00	0.00	0.00	19
leabit	0.27	0.17	0.21	18
leafly	0.27	0.32	0.29	19
leasan	0.25	0.07	0.11	15
lecthr	0.00	0.00	0.00	4
lesgol	0.06	0.05	0.05	20
lesnig	0.38	0.38	0.38	8
lesyel	0.24	0.35	0.29	20
lewwoo	0.00	0.00	0.00	6
linspa	0.10	0.05	0.07	20

```
       lobcur       0.28      0.38      0.32        13
       lobdow       0.33      0.13      0.19        15
       logshr       0.16      0.15      0.15        20
       lotduc       0.22      0.20      0.21        10

     accuracy                           0.18      2473
    macro avg       0.17      0.17      0.16      2473
 weighted avg       0.17      0.18      0.17      2473
```



[48]: '\nTest Accuracy: 0.1678123736352608\n precision recall f1-score support\n\n aldfly 0.08 0.05 0.06 20\n ameavo 0.17 0.12 0.14 8\n amebit 0.00 0.00 0.00 9\n amecro 0.20 0.45 0.27 20\n amegfi 0.20 0.05 0.08 20\n amekes 0.40 0.13 0.20 15\n amepip 0.08 0.10 0.09 20\n amered 0.20 0.10 0.13 20\n amerob 0.19 0.35 0.25 20\n amewig 0.43 0.43 0.43 7\n amewoo 0.50 0.24 0.32 17\n amtspa 0.00 0.00 0.00 15\n annhum 0.08 0.15 0.11

```
20\n      astfly       0.16      0.30      0.21        20\n      baisan
0.00      0.00      0.00        11\n      baleag       0.00      0.00      0.00
7\n      balori       0.19      0.20      0.20        20\n      banswa
0.24      0.20      0.22        20\n      barswa       0.24      0.40      0.30
20\n      bawwar       0.07      0.07      0.07        15\n      belkin1
0.00      0.00      0.00        14\n     belspa2       0.30      0.20      0.24
15\n      bewwre       0.00      0.00      0.00        20\n      bkbcuc
1.00      0.10      0.18        10\n     bkbmag1       0.00      0.00      0.00
12\n      bkbwar       0.07      0.15      0.10        20\n      bkcchi
0.06      0.10      0.07        20\n      bkchum       0.27      0.64      0.38
14\n      bkhgro       0.00      0.00      0.00        20\n      bkpwar
0.29      0.65      0.40        20\n      bktspa       0.25      0.11      0.15
19\n      blkpho       0.25      0.15      0.19        20\n     blugrb1
0.00      0.00      0.00        20\n      blujay       0.05      0.05      0.05
20\n      bnhcow       0.00      0.00      0.00        20\n      boboli
0.19      0.20      0.20        20\n      bongul       0.20      0.25      0.22
8\n      brdowl       0.13      0.10      0.11        20\n      brebla
0.00      0.00      0.00        10\n      brespa       0.09      0.10      0.09
20\n      brncre       0.00      0.00      0.00        20\n      brnthr
0.13      0.20      0.16        20\n      brthum       0.54      0.65      0.59
20\n      brwhaw       0.33      0.07      0.11        15\n      btbwar
0.00      0.00      0.00        15\n      btnwar       0.00      0.00      0.00
20\n      btywar       0.09      0.05      0.07        19\n      buffle
0.00      0.00      0.00         3\n      buggna       0.33      0.15      0.21
20\n      buhvir       0.07      0.10      0.08        20\n      bulori
0.20      0.35      0.25        20\n      bushti       0.15      0.35      0.21
20\n      buwtea       0.29      0.18      0.22        11\n      buwwar
0.15      0.11      0.12        19\n      cacwre       0.18      0.35      0.24
20\n      calgul       0.17      0.12      0.14         8\n      calqua
0.31      0.28      0.29        18\n      camwar       0.14      0.13      0.14
15\n      cangoo       0.31      0.65      0.42        20\n      canwar
0.00      0.00      0.00        20\n      canwre       0.08      0.10      0.09
20\n      carwre       0.00      0.00      0.00        20\n      casfin
0.00      0.00      0.00        15\n     caster1       0.15      0.20      0.17
20\n      casvir       0.22      0.22      0.22        18\n      cedwax
0.14      0.17      0.15        18\n      chispa       0.09      0.10      0.09
20\n      chiswi       0.00      0.00      0.00         6\n      chswar
0.06      0.20      0.09        20\n      chukar       0.00      0.00      0.00
6\n      clanut       0.24      0.44      0.31        16\n      cliswa
0.07      0.08      0.07        13\n      comgol       0.36      0.29      0.32
17\n      comgra       0.06      0.05      0.06        20\n      comloo
0.29      0.15      0.20        13\n      commer       0.19      0.31      0.24
16\n      comnig       0.00      0.00      0.00        17\n      comrav
0.18      0.40      0.25        20\n      comred       0.24      0.25      0.24
20\n      comter       0.34      0.50      0.41        20\n      comyel
0.00      0.00      0.00        20\n      coohaw       0.00      0.00      0.00
18\n      coshum       0.00      0.00      0.00         4\n     cowscj1
```

```
0.00      0.00      0.00        14\n      daejun       0.21      0.30      0.25
20\n      doccor       0.00      0.00      0.00         7\n      dowwoo
0.29      0.10      0.15        20\n      dusfly       0.09      0.10      0.09
20\n      eargre       0.00      0.00      0.00         7\n      easblu
0.00      0.00      0.00        19\n      easkin       0.27      0.19      0.22
16\n      easmea       0.00      0.00      0.00        20\n      easpho
0.00      0.00      0.00        18\n      eastow       0.17      0.10      0.12
20\n      eawpew       0.50      0.10      0.17        20\n      eucdov
0.31      0.50      0.38        20\n      eursta       0.00      0.00      0.00
8\n      evegro       0.02      0.05      0.03        20\n      fiespa
0.20      0.05      0.08        20\n      fiscro       0.10      0.12      0.11
17\n      foxspa       0.14      0.20      0.17        20\n      gadwal
0.10      0.10      0.10        20\n      gcrfin       0.00      0.00      0.00
6\n      gnttow       0.09      0.30      0.13        20\n      gnwtea
0.31      0.55      0.40        20\n      gockin       0.21      0.40      0.28
20\n      gocspa       0.00      0.00      0.00        14\n      goleag
0.00      0.00      0.00        10\n     grbher3       0.00      0.00      0.00
14\n      grcfly       0.14      0.15      0.15        20\n      greegr
0.21      0.15      0.18        20\n      greroa       0.10      0.17      0.13
18\n      greyel       0.40      0.20      0.27        20\n      grhowl
0.24      0.20      0.22        20\n      grnher       0.11      0.08      0.09
13\n      grtgra       0.12      0.05      0.07        20\n      grycat
0.25      0.06      0.10        16\n      gryfly       0.07      0.19      0.10
16\n      haiwoo       0.00      0.00      0.00        20\n      hamfly
0.12      0.15      0.13        20\n      hergul       0.38      0.60      0.46
20\n      herthr       0.16      0.40      0.23        20\n      hoomer
0.00      0.00      0.00         4\n      hoowar       0.11      0.05      0.07
20\n      horgre       0.50      0.15      0.24        13\n      horlar
0.10      0.20      0.13        20\n      houfin       0.23      0.45      0.30
20\n      houspa       0.15      0.40      0.22        20\n      houwre
0.08      0.05      0.06        20\n      indbun       0.33      0.05      0.09
20\n     juntit1       0.33      0.14      0.20        14\n      killde
0.54      0.65      0.59        20\n      labwoo       0.00      0.00      0.00
17\n      larspa       0.00      0.00      0.00        19\n      lazbun
0.19      0.16      0.17        19\n      leabit       0.04      0.06      0.05
18\n      leafly       0.22      0.11      0.14        19\n      leasan
0.20      0.07      0.10        15\n      lecthr       0.00      0.00      0.00
4\n      lesgol       0.10      0.05      0.07        20\n      lesnig
0.44      0.50      0.47         8\n      lesyel       0.29      0.40      0.33
20\n      lewwoo       0.00      0.00      0.00         6\n      linspa
0.00      0.00      0.00        20\n      lobcur       0.12      0.15      0.14
13\n      lobdow       0.00      0.00      0.00        15\n      logshr
0.11      0.10      0.11        20\n      lotduc       0.50      0.40      0.44
10\n\n    accuracy                           0.17      2473\n   macro avg
0.15      0.16      0.14      2473\nweighted avg       0.16      0.17      0.15
2473\n\n/usr/local/lib/python3.11/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
```

```
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.\n
_warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))\n/usr/local/lib/python3.11/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.\n
_warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))\n/usr/local/lib/python3.11/dist-
packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning:
Precision is ill-defined and being set to 0.0 in labels with no predicted
samples. Use `zero_division` parameter to control this behavior.\n
_warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))\n\n'
```

1.11 Critical Discussion & Limitations

Several factors constrained performance. First, class imbalance meant the network saw scant examples of rare species, biasing it toward abundant classes. Second, we never trained directly on raw waveforms or spectrograms, potentially discarding discriminative patterns present in time-frequency images. Third, our evaluation relied on a single train-test split, so variance across different splits is unknown; a five-fold cross-validation would deliver more reliable metrics. Finally, our models treated each recording independently, ignoring contextual cues such as geolocation or seasonality that experts use when identifying birds.

1.12 Future Work

Future iterations should implement stratified k-fold cross-validation, experiment with convolutional or recurrent layers on spectrogram inputs, and apply data-balancing strategies such as focal loss or SMOTE. Ensemble methods (averaging predictions from complementary architectures) might also close the gap to state-of-the-art performance. Lastly, integrating metadata (e.g., GPS tags, time-of-year) could give the classifier ecological context and further improve accuracy.

1.13 Conclusion

Our study shows that even modest models can extract informative patterns from condensed audio features. Starting from a 12% baseline, systematic hyperparameter optimisation and a switch to PyTorch culminated in 17% accuracy. While there remains ample room before we rival Kaggle leaders, the project met its educational objectives: it deepened our fluency in audio feature engineering, Bayesian hyperparameter search, and comparative model evaluation, skills readily transferable to broader machine-learning challenges.